RISPERDAL- risperidone tablet
RISPERDAL M-TAB- risperidone tablet, orally disintegrating
RISPERDAL- risperidone solution
Janssen Pharmaceuticals, Inc.

HIGHLIGHTS OF PRESCRIBING INFORMATION
These highlights do not include all the information needed to use RISPERDAL® safely and effectively. See full prescribing information for RISPERDAL®.
RISPERDAL® (risperidone) tablets, for oral use
RISPERDAL® (risperidone) oral solution
RISPERDAL® M-TAB® (risperidone) orally disintegrating tablets
Initial U.S. Approval: 1993

WARNING: INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS

See full prescribing information for complete boxed warning.

Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death. RISPERDAL is not approved for use in patients with dementia-related psychosis. (5.1)

RECENT MAJOR CHANGES

Warnings and Precautions (5.6)	1/2025

INDICATIONS AND USAGE

RISPERDAL is an atypical antipsychotic indicated for:

- Treatment of schizophrenia (1.1)
- As monotherapy or adjunctive therapy with lithium or valproate, for the treatment of acute manic or mixed episodes associated with Bipolar I Disorder (1.2)
- Treatment of irritability associated with autistic disorder (1.3)

DOSAGE AND ADMINISTRATION

- Recommended daily dosage:

	Initial Dose	Target Dose	Effective Dose Range
Schizophrenia: adults (2.1)	2 mg	4 to 8 mg	4 to 16 mg
Schizophrenia: adolescents (2.1)	0.5 mg	3 mg	1 to 6 mg
Bipolar mania: adults (2.2)	2 to 3 mg	1 to 6 mg	1 to 6 mg
Bipolar mania: in children and adolescents (2.2)	0.5 mg	1 to 2.5 mg	1 to 6 mg
Irritability associated with autistic disorder (2.3)	0.25 mg (Weight < 20 kg) 0.5 mg (Weight ≥20 kg)	0.5 mg (<20 kg) 1 mg (≥20 kg)	0.5 to 3 mg

- Severe Renal or Hepatic Impairment in Adults: Use a lower starting dose of 0.5 mg twice daily. May increase to dosages above 1.5 mg twice daily at intervals of at least one week. (2.4)
- Oral Solution: Can be administered directly from calibrated oral dosing syringe or mixed with beverage (water, coffee, orange juice, or low-fat milk). (2.6)

- M-TAB Orally Disintegrating Tablets: Open the blister only when ready to administer, and immediately place tablet on the tongue. Can be swallowed with or without liquid. (2.7)

DOSAGE FORMS AND STRENGTHS

- Tablets: 0.5 mg, 1 mg, 2 mg, 3 mg, and 4 mg (3)
- Oral solution: 1 mg per mL (3)
- Orally disintegrating tablets: 0.5 mg, 1 mg, 2 mg, 3 mg, and 4 mg (3)

CONTRAINDICATIONS

- Known hypersensitivity to risperidone, paliperidone, or to any excipients in RISPERDAL. (4)

WARNINGS AND PRECAUTIONS

- Cerebrovascular events, including stroke, in elderly patients with dementia-related psychosis: RISPERDAL is not approved for use in patients with dementia-related psychosis. (5.2)
- Neuroleptic Malignant Syndrome: Manage with immediate discontinuation of RISPERDAL and close monitoring. (5.3)
- Tardive dyskinesia: Consider discontinuing RISPERDAL if clinically indicated. (5.4)
- Metabolic Changes: Atypical antipsychotic drugs have been associated with metabolic changes that may increase cardiovascular/ cerebrovascular risk. These metabolic changes include hyperglycemia, dyslipidemia, and weight gain. (5.5)
  - *Hyperglycemia and Diabetes Mellitus:* Monitor patients for symptoms of hyperglycemia including polydipsia, polyuria, polyphagia, and weakness. Monitor glucose regularly in patients with diabetes or at risk for diabetes. (5.5)
  - *Dyslipidemia:* Undesirable alterations have been observed in patients treated with atypical antipsychotics. (5.5)
  - *Weight Gain:* Significant weight gain has been reported. Monitor weight gain. (5.5)
- Hyperprolactinemia: Prolactin elevations occur and persist during chronic administration. (5.6)
- Orthostatic hypotension: For patients at risk, consider a lower starting dose and slower titration. (5.7)
- Leukopenia, Neutropenia, and Agranulocytosis: Perform complete blood counts in patients with a history of clinically significant low white blood cell count (WBC). Consider discontinuing RISPERDAL if a clinically significant decline in WBC occurs in the absence of other causative factors. (5.9)
- Potential for cognitive and motor impairment: Use caution when operating machinery. (5.10)
- Seizures: Use cautiously in patients with a history of seizures or with conditions that lower the seizure threshold. (5.11)

ADVERSE REACTIONS

The most common adverse reactions in clinical trials (≥5% and twice placebo) were parkinsonism, akathisia, dystonia, tremor, sedation, dizziness, anxiety, blurred vision, nausea, vomiting, upper abdominal pain, stomach discomfort, dyspepsia, diarrhea, salivary hypersecretion, constipation, dry mouth, increased appetite, increased weight, fatigue, rash, nasal congestion, upper respiratory tract infection, nasopharyngitis, and pharyngolaryngeal pain. (6)

To report SUSPECTED ADVERSE REACTIONS, contact Janssen Pharmaceuticals, Inc. at 1-800-526-7736 or FDA at 1-800-FDA-1088 or *www.fda.gov/medwatch*

DRUG INTERACTIONS

- Carbamazepine and other enzyme inducers decrease plasma concentrations of risperidone. Increase the RISPERDAL dose up to double the patient's usual dose. Titrate slowly. (7.1)
- Fluoxetine, paroxetine, and other CYP 2D6 enzyme inhibitors increase plasma concentrations of risperidone. Reduce the initial dose. Do not exceed a final dose of 8 mg per day of RISPERDAL. (7.1)

USE IN SPECIFIC POPULATIONS

- Pregnancy: May cause extrapyramidal and/or withdrawal symptoms in neonates with third trimester exposure. (8.1)

See 17 for PATIENT COUNSELING INFORMATION.

Revised: 5/2026

FULL PRESCRIBING INFORMATION: CONTENTS*

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FULL PRESCRIBING INFORMATION

WARNING: INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS

Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death. RISPERDAL is not approved for the treatment of patients with dementia-related psychosis. *[see Warnings and Precautions (5.1)]*

1 INDICATIONS AND USAGE

1.1 Schizophrenia

RISPERDAL (risperidone) is indicated for the treatment of schizophrenia. Efficacy was established in 4 short-term trials in adults, 2 short-term trials in adolescents (ages 13 to 17 years), and one long-term maintenance trial in adults *[see Clinical Studies (14.1)]*.

1.2 Bipolar Mania

Monotherapy

RISPERDAL is indicated for the treatment of acute manic or mixed episodes associated with Bipolar I Disorder. Efficacy was established in 2 short-term trials in adults and one short-term trial in children and adolescents (ages 10 to 17 years) *[see Clinical Studies (14.2)]*.

Adjunctive Therapy

RISPERDAL adjunctive therapy with lithium or valproate is indicated for the treatment of acute manic or mixed episodes associated with Bipolar I Disorder. Efficacy was established in one short-term trial in adults *[see Clinical Studies (14.3)]*.

1.3 Irritability Associated with Autistic Disorder

RISPERDAL is indicated for the treatment of irritability associated with autistic disorder, including symptoms of aggression towards others, deliberate self-injuriousness, temper tantrums, and quickly changing moods. Efficacy was established in 3 short-term trials in children and adolescents (ages 5 to 17 years) *[see Clinical Studies (14.4)]*.

2 DOSAGE AND ADMINISTRATION

Table 1. Recommended Daily Dosage by Indication

	Initial Dose	Titration (Increments)	Target Dose	Effective Dose Range
Schizophrenia: adults (2.1)	2 mg	1 to 2 mg	4 to 8 mg	4 to 16 mg
Schizophrenia: adolescents (2.1)	0.5 mg	0.5 to 1 mg	3 mg	1 to 6 mg
Bipolar mania: adults (2.2)	2 to 3 mg	1 mg	1 to 6 mg	1 to 6 mg
Bipolar mania: children and adolescents (2.2)	0.5 mg	0.5 to 1 mg	1 to 2.5 mg	1 to 6 mg
Irritability in autistic disorder (2.3)	0.25 mg Can increase to 0.5 mg by Day 4: (body weight less than 20 kg) 0.5 mg Can increase to 1 mg by	After Day 4, at intervals of > 2 weeks: 0.25 mg (body weight less than 20 kg) 0.5 mg (body weight	0.5 mg: (body weight less than 20 kg) 1 mg: (body weight greater	0.5 to 3 mg

	Day 4: (body weight greater than or equal to 20 kg)	greater than or equal to 20 kg)	than or equal to 20 kg)	

Severe Renal and Hepatic Impairment in Adults: use a lower starting dose of 0.5 mg twice daily. May increase to dosages above 1.5 mg twice daily at intervals of one week or longer.

2.1 Schizophrenia

Adults

Usual Initial Dose

RISPERDAL can be administered once or twice daily. Initial dosing is 2 mg per day. May increase the dose at intervals of 24 hours or greater, in increments of 1 to 2 mg per day, as tolerated, to a recommended dose of 4 to 8 mg per day. In some patients, slower titration may be appropriate. Efficacy has been demonstrated in a range of 4 mg to 16 mg per day. However, doses above 6 mg per day for twice daily dosing were not demonstrated to be more efficacious than lower doses, were associated with more extrapyramidal symptoms and other adverse effects, and are generally not recommended. In a single study supporting once-daily dosing, the efficacy results were generally stronger for 8 mg than for 4 mg. The safety of doses above 16 mg per day has not been evaluated in clinical trials *[see Clinical Studies (14.1)]*.

Adolescents

The initial dose is 0.5 mg once daily, administered as a single-daily dose in the morning or evening. The dose may be adjusted at intervals of 24 hours or greater, in increments of 0.5 mg or 1 mg per day, as tolerated, to a recommended dose of 3 mg per day. Although efficacy has been demonstrated in studies of adolescent patients with schizophrenia at doses between 1 mg to 6 mg per day, no additional benefit was observed above 3 mg per day, and higher doses were associated with more adverse events. Doses higher than 6 mg per day have not been studied.

Patients experiencing persistent somnolence may benefit from administering half the daily dose twice daily.

Maintenance Therapy

While it is unknown how long a patient with schizophrenia should remain on RISPERDAL, the effectiveness of RISPERDAL 2 mg per day to 8 mg per day at delaying relapse was demonstrated in a controlled trial in adult patients who had been clinically stable for at least 4 weeks and were then followed for a period of 1 to 2 years *[see Clinical Studies (14.1)]*. Both adult and adolescent patients who respond acutely should generally be maintained on their effective dose beyond the acute episode. Patients should be periodically reassessed to determine the need for maintenance treatment.

Reinitiation of Treatment in Patients Previously Discontinued

Although there are no data to specifically address reinitiation of treatment, it is recommended that after an interval off RISPERDAL, the initial titration schedule should be followed.

Switching From Other Antipsychotics

There are no systematically collected data to specifically address switching schizophrenic patients from other antipsychotics to RISPERDAL, or treating patients with concomitant antipsychotics.

2.2 Bipolar Mania

Usual Dose

Adults

The initial dose range is 2 mg to 3 mg per day. The dose may be adjusted at intervals of 24 hours or greater, in increments of 1 mg per day. The effective dose range is 1 mg to 6 mg per day, as studied in the short-term, placebo-controlled trials. In these trials, short-term (3 week) anti-manic efficacy was demonstrated in a flexible dosage range of 1 mg to 6 mg per day *[see Clinical Studies (14.2, 14.3)]*. RISPERDAL doses higher than 6 mg per day were not studied.

Pediatrics

The initial dose is 0.5 mg once daily, administered as a single-daily dose in the morning or evening. The dose may be adjusted at intervals of 24 hours or greater, in increments of 0.5 mg or 1 mg per day, as tolerated, to the recommended target dose of 1 mg to 2.5 mg per day. Although efficacy has been demonstrated in studies of pediatric patients with bipolar mania at doses between 0.5 mg and 6 mg per day, no additional benefit was observed above 2.5 mg per day, and higher doses were associated with more adverse events. Doses higher than 6 mg per day have not been studied.

Patients experiencing persistent somnolence may benefit from administering half the daily dose twice daily.

Maintenance Therapy

There is no body of evidence available from controlled trials to guide a clinician in the longer-term management of a patient who improves during treatment of an acute manic episode with RISPERDAL. While it is generally agreed that pharmacological treatment beyond an acute response in mania is desirable, both for maintenance of the initial response and for prevention of new manic episodes, there are no systematically obtained data to support the use of RISPERDAL in such longer-term treatment (i.e., beyond 3 weeks). The physician who elects to use RISPERDAL for extended periods should periodically re-evaluate the long-term risks and benefits of the drug for the individual patient.

2.3 Irritability Associated with Autistic Disorder – Pediatrics (Children and Adolescents)

The dosage of RISPERDAL should be individualized according to the response and tolerability of the patient. The total daily dose of RISPERDAL can be administered once daily, or half the total daily dose can be administered twice daily.

For patients with body weight less than 20 kg, initiate dosing at 0.25 mg per day. For

patients with body weight greater than or equal to 20 kg, initiate dosing at 0.5 mg per day. After a minimum of four days, the dose may be increased to the recommended dose of 0.5 mg per day for patients less than 20 kg and 1.0 mg per day for patients greater than or equal to 20 kg. Maintain this dose for a minimum of 14 days. In patients not achieving sufficient clinical response, the dose may be increased at intervals of 2 weeks or greater, in increments of 0.25 mg per day for patients less than 20 kg, or increments of 0.5 mg per day for patients greater than or equal to 20 kg. The effective dose range is 0.5 mg to 3 mg per day. No dosing data are available for children who weigh less than 15 kg.

Once sufficient clinical response has been achieved and maintained, consider gradually lowering the dose to achieve the optimal balance of efficacy and safety. The physician who elects to use RISPERDAL for extended periods should periodically re-evaluate the long-term risks and benefits of the drug for the individual patient.

Patients experiencing persistent somnolence may benefit from a once-daily dose administered at bedtime or administering half the daily dose twice daily, or a reduction of the dose.

2.4 Dosing in Patients with Severe Renal or Hepatic Impairment

For patients with severe renal impairment (Clcr< 30 mL/min) or hepatic impairment (10–15 points on Child Pugh System), the initial starting dose is 0.5 mg twice daily. The dose may be increased in increments of 0.5 mg or less, administered twice daily. For doses above 1.5 mg twice daily, increase in intervals of one week or greater *[see Use in Specific Populations (8.6 and 8.7)].*

2.5 Dose Adjustments for Specific Drug Interactions

When RISPERDAL is co-administered with enzyme inducers (e.g., carbamazepine), the dose of RISPERDAL should be increased up to double the patient's usual dose. It may be necessary to decrease the RISPERDAL dose when enzyme inducers such as carbamazepine are discontinued *[see Drug Interactions (7.1)]*. Similar effect may be expected with co-administration of RISPERDAL with other enzyme inducers (e.g., phenytoin, rifampin, and phenobarbital).

When fluoxetine or paroxetine is co-administered with RISPERDAL, the dose of RISPERDAL should be reduced. The RISPERDAL dose should not exceed 8 mg per day in adults when co-administered with these drugs. When initiating therapy, RISPERDAL should be titrated slowly. It may be necessary to increase the RISPERDAL dose when enzyme inhibitors such as fluoxetine or paroxetine are discontinued *[see Drug Interactions (7.1)]*.

2.6 Administration of RISPERDAL Oral Solution

RISPERDAL Oral Solution can be administered directly from the calibrated oral dosing syringe, or can be mixed with a beverage prior to administration. RISPERDAL Oral Solution is compatible in the following beverages: water, coffee, orange juice, and low-fat milk; it is NOT compatible with either cola or tea.

2.7 Directions for Use of RISPERDAL M-TAB Orally Disintegrating Tablets

Tablet Accessing

RISPERDAL M-TAB Orally Disintegrating Tablets 0.5 mg, 1 mg, and 2 mg

RISPERDAL M-TAB Orally Disintegrating Tablets 0.5 mg, 1 mg, and 2 mg are supplied in blister packs of 4 tablets each.

Do not open the blister until ready to administer. For single tablet removal, separate one of the four blister units by tearing apart at the perforations. Bend the corner where indicated. Peel back foil to expose the tablet. DO NOT push the tablet through the foil because this could damage the tablet.

RISPERDAL M-TAB Orally Disintegrating Tablets 3 mg and 4 mg

RISPERDAL M-TAB Orally Disintegrating Tablets 3 mg and 4 mg are supplied in a child-resistant pouch containing a blister with 1 tablet each.

The child-resistant pouch should be torn open at the notch to access the blister. Do not open the blister until ready to administer. Peel back foil from the side to expose the tablet. DO NOT push the tablet through the foil, because this could damage the tablet.

Tablet Administration

Using dry hands, remove the tablet from the blister unit and immediately place the entire RISPERDAL M-TAB Orally Disintegrating Tablet on the tongue. The RISPERDAL M-TAB Orally Disintegrating Tablet should be consumed immediately, as the tablet cannot be stored once removed from the blister unit. RISPERDAL M-TAB Orally Disintegrating Tablets disintegrate in the mouth within seconds and can be swallowed subsequently with or without liquid. Patients should not attempt to split or to chew the tablet.

3 DOSAGE FORMS AND STRENGTHS

RISPERDAL Tablets are available in the following strengths and colors: 0.5 mg (red-brown), 1 mg (white), 2 mg (orange), 3 mg (yellow), and 4 mg (green). All are capsule shaped, and imprinted with "JANSSEN" on one side and either "Ris 0.5", "R1", "R2", "R3", or "R4" on the other side according to their respective strengths.

RISPERDAL Oral Solution is available in a 1 mg/mL strength.

RISPERDAL M-TAB Orally Disintegrating Tablets are available in the following strengths, colors, and shapes: 0.5 mg (light coral, round), 1 mg (light coral, square), 2 mg (coral, square), 3 mg (coral, round), and 4 mg (coral, round). All are biconvex and etched on one side with "R0.5", "R1", "R2", "R3", or "R4" according to their respective strengths.

4 CONTRAINDICATIONS

RISPERDAL is contraindicated in patients with a known hypersensitivity to either risperidone or paliperidone, or to any of the excipients in the RISPERDAL formulation. Hypersensitivity reactions, including anaphylactic reactions and angioedema, have been reported in patients treated with risperidone and in patients treated with paliperidone. Paliperidone is a metabolite of risperidone.

5 WARNINGS AND PRECAUTIONS

5.1 Increased Mortality in Elderly Patients with Dementia-Related Psychosis

Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death. Analyses of 17 placebo-controlled trials (modal duration of 10 weeks), largely in patients taking atypical antipsychotic drugs, revealed a risk of death in drug-treated patients of between 1.6 to 1.7 times the risk of death in placebo-treated patients. Over the course of a typical 10-week controlled trial, the rate of death in drug-treated patients was about 4.5%, compared to a rate of about 2.6% in the placebo group. Although the causes of death were varied, most of the deaths appeared to be either cardiovascular (e.g., heart failure, sudden death) or infectious (e.g., pneumonia) in nature. Observational studies suggest that, similar to atypical antipsychotic drugs, treatment with conventional antipsychotic drugs may increase mortality. The extent to which the findings of increased mortality in observational studies may be attributed to the antipsychotic drug as opposed to some characteristic(s) of the patients is not clear.

In two of four placebo-controlled trials in elderly patients with dementia-related psychosis, a higher incidence of mortality was observed in patients treated with furosemide plus RISPERDAL when compared to patients treated with RISPERDAL alone or with placebo plus furosemide. No pathological mechanism has been identified to explain this finding, and no consistent pattern for cause of death was observed.

RISPERDAL (risperidone) is not approved for the treatment of dementia-related psychosis *[see Boxed Warning]*.

5.2 Cerebrovascular Adverse Reactions, Including Stroke, in Elderly Patients with Dementia-Related Psychosis

Cerebrovascular adverse reactions (e.g., stroke, transient ischemic attack), including fatalities, were reported in patients (mean age 85 years; range 73–97) in trials of risperidone in elderly patients with dementia-related psychosis. In placebo-controlled trials, there was a significantly higher incidence of cerebrovascular adverse events in patients treated with risperidone compared to patients treated with placebo. RISPERDAL is not approved for the treatment of patients with dementia-related psychosis. *[see Boxed Warning and Warnings and Precautions (5.1)].*

5.3 Neuroleptic Malignant Syndrome

Neuroleptic Malignant Syndrome (NMS), a potentially fatal symptom complex, has been reported in association with antipsychotic drugs. Clinical manifestations of NMS are hyperpyrexia, muscle rigidity, altered mental status including delirium, and autonomic instability (irregular pulse or blood pressure, tachycardia, diaphoresis, and cardiac dysrhythmia). Additional signs may include elevated creatine phosphokinase, myoglobinuria (rhabdomyolysis), and acute renal failure.

If NMS is suspected, immediately discontinue RISPERDAL and provide symptomatic treatment and monitoring.

5.4 Tardive Dyskinesia

Tardive dyskinesia, a syndrome consisting of potentially irreversible, involuntary, dyskinetic movements, may develop in patients treated with antipsychotic drugs. Although the prevalence of the syndrome appears to be highest among the elderly, especially elderly women, it is impossible to predict which patients will develop the syndrome. Whether antipsychotic drug products differ in their potential to cause tardive

dyskinesia is unknown.

The risk of developing tardive dyskinesia and the likelihood that it will become irreversible increase with the duration of treatment and the cumulative dose. The syndrome can develop after relatively brief treatment periods, even at low doses. It may also occur after discontinuation of treatment.

Tardive dyskinesia may remit, partially or completely, if antipsychotic treatment is discontinued. Antipsychotic treatment, itself, however, may suppress (or partially suppress) the signs and symptoms of the syndrome, possibly masking the underlying process. The effect that symptomatic suppression has upon the long-term course of the syndrome is unknown.

Given these considerations, RISPERDAL should be prescribed in a manner that is most likely to minimize the occurrence of tardive dyskinesia. Chronic antipsychotic treatment should generally be reserved for patients: (1) who suffer from a chronic illness that is known to respond to antipsychotic drugs, and (2) for whom alternative, equally effective, but potentially less harmful treatments are not available or appropriate. In patients who do require chronic treatment, use the lowest dose and the shortest duration of treatment producing a satisfactory clinical response. Periodically reassess the need for continued treatment.

If signs and symptoms of tardive dyskinesia appear in a patient on RISPERDAL, drug discontinuation should be considered. However, some patients may require treatment with RISPERDAL despite the presence of the syndrome.

5.5 Metabolic Changes

Atypical antipsychotic drugs have been associated with metabolic changes that may increase cardiovascular/cerebrovascular risk. These metabolic changes include hyperglycemia, dyslipidemia, and body weight gain. While all of the drugs in the class have been shown to produce some metabolic changes, each drug has its own specific risk profile.

Hyperglycemia and Diabetes Mellitus

Hyperglycemia and diabetes mellitus, in some cases extreme and associated with ketoacidosis or hyperosmolar coma or death, have been reported in patients treated with atypical antipsychotics including RISPERDAL. Assessment of the relationship between atypical antipsychotic use and glucose abnormalities is complicated by the possibility of an increased background risk of diabetes mellitus in patients with schizophrenia and the increasing incidence of diabetes mellitus in the general population. Given these confounders, the relationship between atypical antipsychotic use and hyperglycemia-related adverse events is not completely understood. However, epidemiological studies suggest an increased risk of treatment-emergent hyperglycemia-related adverse events in patients treated with the atypical antipsychotics. Precise risk estimates for hyperglycemia-related adverse events in patients treated with atypical antipsychotics are not available.

Patients with an established diagnosis of diabetes mellitus who are started on atypical antipsychotics, including RISPERDAL, should be monitored regularly for worsening of glucose control. Patients with risk factors for diabetes mellitus (e.g., obesity, family history of diabetes) who are starting treatment with atypical antipsychotics, including RISPERDAL, should undergo fasting blood glucose testing at the beginning of treatment

and periodically during treatment. Any patient treated with atypical antipsychotics, including RISPERDAL, should be monitored for symptoms of hyperglycemia including polydipsia, polyuria, polyphagia, and weakness. Patients who develop symptoms of hyperglycemia during treatment with atypical antipsychotics, including RISPERDAL, should undergo fasting blood glucose testing. In some cases, hyperglycemia has resolved when the atypical antipsychotic, including RISPERDAL, was discontinued; however, some patients required continuation of anti-diabetic treatment despite discontinuation of RISPERDAL.

Pooled data from three double-blind, placebo-controlled schizophrenia studies and four double-blind, placebo-controlled bipolar monotherapy studies are presented in Table 2.

Table 2. Change in Random Glucose from Seven Placebo-Controlled, 3- to 8-Week, Fixed- or Flexible-Dose Studies in Adult Subjects with Schizophrenia or Bipolar Mania

		RISPERDAL	
	Placebo	**1–8 mg/day**	**>8–16 mg/day**
	Mean change from baseline (mg/dL)		
	n=555	**n=748**	**n=164**
Serum Glucose	-1.4	0.8	0.6
	Proportion of patients with shifts		
Serum Glucose (<140 mg/dL to ≥200 mg/dL)	0.6% (3/525)	0.4% (3/702)	0% (0/158)

In longer-term, controlled and uncontrolled studies, RISPERDAL was associated with a mean change in glucose of +2.8 mg/dL at Week 24 (n=151) and +4.1 mg/dL at Week 48 (n=50).

Data from the placebo-controlled 3- to 6-week study in children and adolescents with schizophrenia (13–17 years of age), bipolar mania (10–17 years of age), or autistic disorder (5 to 17 years of age) are presented in Table 3.

Table 3. Change in Fasting Glucose from Three Placebo-Controlled, 3- to 6-Week, Fixed-Dose Studies in Children and Adolescents with Schizophrenia (13–17 years of age), Bipolar Mania (10–17 years of age), or Autistic Disorder (5 to 17 years of age)

		RISPERDAL
	Placebo	**0.5–6 mg/day**
	Mean change from baseline (mg/dL)	
	n=76	**n=135**
Serum Glucose	-1.3	2.6
	Proportion of patients with shifts	
Serum Glucose (<100 mg/dL to ≥126 mg/dL)	0% (0/64)	0.8% (1/120)

In longer-term, uncontrolled, open-label extension pediatric studies, RISPERDAL was associated with a mean change in fasting glucose of +5.2 mg/dL at Week 24 (n=119).

Dyslipidemia

Undesirable alterations in lipids have been observed in patients treated with atypical antipsychotics.

Pooled data from 7 placebo-controlled, 3- to 8- week, fixed- or flexible-dose studies in adult subjects with schizophrenia or bipolar mania are presented in Table 4.

Table 4. Change in Random Lipids from Seven Placebo-Controlled, 3- to 8-Week, Fixed- or Flexible-Dose Studies in Adult Subjects with Schizophrenia or Bipolar Mania

		RISPERDAL	
	Placebo	**1–8 mg/day**	**>8–16 mg/day**
	Mean change from baseline (mg/dL)		
Cholesterol	**n=559**	**n=742**	**n=156**
Change from baseline	0.6	6.9	1.8
Triglycerides	**n=183**	**n=307**	**n=123**
Change from baseline	-17.4	-4.9	-8.3
	Proportion of patients With Shifts		
Cholesterol (<200 mg/dL to ≥240 mg/dL)	2.7% (10/368)	4.3% (22/516)	6.3% (6/96)
Triglycerides (<500 mg/dL to ≥500 mg/dL)	1.1% (2/180)	2.7% (8/301)	2.5% (3/121)

In longer-term, controlled and uncontrolled studies, RISPERDAL was associated with a mean change in (a) non-fasting cholesterol of +4.4 mg/dL at Week 24 (n=231) and +5.5 mg/dL at Week 48 (n=86); and (b) non-fasting triglycerides of +19.9 mg/dL at Week 24 (n=52).

Pooled data from 3 placebo-controlled, 3- to 6-week, fixed-dose studies in children and adolescents with schizophrenia (13–17 years of age), bipolar mania (10–17 years of age), or autistic disorder (5–17 years of age) are presented in Table 5.

Table 5. Change in Fasting Lipids from Three Placebo-Controlled, 3- to 6-Week, Fixed-Dose Studies in Children and Adolescents with Schizophrenia (13–17 Years of Age), Bipolar Mania (10–17 Years of Age), or Autistic Disorder (5 to 17 Years of Age)

		RISPERDAL
	Placebo	**0.5–6 mg/day**
	Mean change from baseline (mg/dL)	
Cholesterol	**n=74**	**n=133**
Change from baseline	0.3	-0.3

LDL	**n=22**	**n=22**
Change from baseline	3.7	0.5
HDL	**n=22**	**n=22**
Change from baseline	1.6	-1.9
Triglycerides	**n=77**	**n=138**
Change from baseline	-9.0	-2.6
	Proportion of patients with shifts	
Cholesterol (<170 mg/dL to ≥200 mg/dL)	2.4% (1/42)	3.8% (3/80)
LDL (<110 mg/dL to ≥130 mg/dL)	0% (0/16)	0% (0/16)
HDL (≥40 mg/dL to <40 mg/dL)	0% (0/19)	10% (2/20)
Triglycerides (<150 mg/dL to ≥200 mg/dL)	1.5% (1/65)	7.1% (8/113)

In longer-term, uncontrolled, open-label extension pediatric studies, RISPERDAL was associated with a mean change in (a) fasting cholesterol of +2.1 mg/dL at Week 24 (n=114); (b) fasting LDL of -0.2 mg/dL at Week 24 (n=103); (c) fasting HDL of +0.4 mg/dL at Week 24 (n=103); and (d) fasting triglycerides of +6.8 mg/dL at Week 24 (n=120).

Weight Gain

Weight gain has been observed with atypical antipsychotic use. Clinical monitoring of weight is recommended.

Data on mean changes in body weight and the proportion of subjects meeting a weight gain criterion of 7% or greater of body weight from 7 placebo-controlled, 3- to 8- week, fixed- or flexible-dose studies in adult subjects with schizophrenia or bipolar mania are presented in Table 6.

Table 6. Mean Change in Body Weight (kg) and the Proportion of Subjects with ≥7% Gain in Body Weight From Seven Placebo-Controlled, 3- to 8-Week, Fixed- or Flexible-Dose Studies in Adult Subjects With Schizophrenia or Bipolar Mania

		RISPERDAL	
	Placebo (n=597)	**1–8 mg/day (n=769)**	**>8–16 mg/day (n=158)**
Weight (kg)			
Change from baseline	-0.3	0.7	2.2
Weight Gain			
≥7% increase from baseline	2.9%	8.7%	20.9%

In longer-term, controlled and uncontrolled studies, RISPERDAL was associated with a mean change in weight of +4.3 kg at Week 24 (n=395) and +5.3 kg at Week 48 (n=203).

Data on mean changes in body weight and the proportion of subjects meeting the criterion of ≥7% gain in body weight from nine placebo-controlled, 3- to 8-week, fixed-dose studies in children and adolescents with schizophrenia (13–17 years of age), bipolar mania (10–17 years of age), autistic disorder (5–17 years of age), or other psychiatric disorders (5–17 years of age) are presented in Table 7.

Table 7. Mean Change in Body Weight (kg) and the Proportion of Subjects With ≥7% Gain in Body Weight From Nine Placebo-Controlled, 3- to 8-Week, Fixed-Dose Studies in Children and Adolescents With Schizophrenia (13–17 Years of Age), Bipolar Mania (10–17 Years of Age), Autistic Disorder (5 to 17 Years of Age) or Other Psychiatric Disorders (5–17 Years of Age)

	Placebo (n=375)	RISPERDAL 0.5–6 mg/day (n=448)
Weight (kg)		
Change from baseline	0.6	2.0
Weight Gain		
≥7% increase from baseline	6.9%	32.6%

In longer-term, uncontrolled, open-label extension pediatric studies, RISPERDAL was associated with a mean change in weight of +5.5 kg at Week 24 (n=748) and +8.0 kg at Week 48 (n=242).

In a long-term, open-label extension study in adolescent patients with schizophrenia, weight increase was reported as a treatment-emergent adverse event in 14% of patients. In 103 adolescent patients with schizophrenia, a mean increase of 9.0 kg was observed after 8 months of RISPERDAL treatment. The majority of that increase was observed within the first 6 months. The average percentiles at baseline and 8 months, respectively, were 56 and 72 for weight, 55 and 58 for height, and 51 and 71 for body mass index.

In long-term, open-label trials (studies in patients with autistic disorder or other psychiatric disorders), a mean increase of 7.5 kg after 12 months of RISPERDAL treatment was observed, which was higher than the expected normal weight gain (approximately 3 to 3.5 kg per year adjusted for age, based on Centers for Disease Control and Prevention normative data). The majority of that increase occurred within the first 6 months of exposure to RISPERDAL. The average percentiles at baseline and 12 months, respectively, were 49 and 60 for weight, 48 and 53 for height, and 50 and 62 for body mass index.

In one 3-week, placebo-controlled trial in children and adolescent patients with acute manic or mixed episodes of bipolar I disorder, increases in body weight were higher in the RISPERDAL groups than the placebo group, but not dose related (1.90 kg in the RISPERDAL 0.5–2.5 mg group, 1.44 kg in the RISPERDAL 3–6 mg group, and 0.65 kg in the placebo group). A similar trend was observed in the mean change from baseline in body mass index.

When treating pediatric patients with RISPERDAL for any indication, weight gain should be assessed against that expected with normal growth.

5.6 Hyperprolactinemia

As with other drugs that antagonize dopamine D_2 receptors, RISPERDAL elevates prolactin levels and the elevation persists during chronic administration. RISPERDAL is associated with higher levels of prolactin elevation than other antipsychotic agents.

Hyperprolactinemia may suppress hypothalamic GnRH, resulting in reduced pituitary gonadotropin secretion. This, in turn, may inhibit reproductive function by impairing gonadal steroidogenesis in both female and male patients. Galactorrhea, amenorrhea, gynecomastia, and impotence have been reported in patients receiving prolactin-elevating compounds. Long-standing hyperprolactinemia when associated with hypogonadism may lead to decreased bone density in both female and male subjects.

Tissue culture experiments indicate that approximately one-third of human breast cancers are prolactin dependent *in vitro*, a factor of potential importance if the prescription of these drugs is contemplated in a patient with previously detected breast cancer. An increase in pituitary gland, mammary gland, and pancreatic islet cell neoplasia (mammary adenocarcinomas, pituitary and pancreatic adenomas) was observed in the risperidone carcinogenicity studies conducted in mice and rats *[see Nonclinical Toxicology (13.1)]*. Published epidemiologic studies have shown inconsistent results when exploring the potential association between hyperprolactinemia and breast cancer.

5.7 Orthostatic Hypotension

RISPERDAL may induce orthostatic hypotension associated with dizziness, tachycardia, and in some patients, syncope, especially during the initial dose-titration period, probably reflecting its alpha-adrenergic antagonistic properties. Syncope was reported in 0.2% (6/2607) of RISPERDAL-treated patients in Phase 2 and 3 studies in adults with schizophrenia. The risk of orthostatic hypotension and syncope may be minimized by limiting the initial dose to 2 mg total (either once daily or 1 mg twice daily) in normal adults and 0.5 mg twice daily in the elderly and patients with renal or hepatic impairment *[see Dosage and Administration (2.1, 2.4)]*. Monitoring of orthostatic vital signs should be considered in patients for whom this is of concern. A dose reduction should be considered if hypotension occurs. RISPERDAL should be used with particular caution in patients with known cardiovascular disease (history of myocardial infarction or ischemia, heart failure, or conduction abnormalities), cerebrovascular disease, and conditions which would predispose patients to hypotension, e.g., dehydration and hypovolemia, and in the elderly and patients with renal or hepatic impairment. Monitoring of orthostatic vital signs should be considered if hypotension occurs. Clinically significant hypotension has been observed with concomitant use of RISPERDAL and antihypertensive medication.

5.8 Falls

Somnolence, postural hypotension, motor and sensory instability have been reported with the use of antipsychotics, including RISPERDAL, which may lead to falls and, consequently, fractures or other fall-related injuries. For patients, particularly the elderly, with diseases, conditions, or medications that could exacerbate these effects, assess the risk of falls when initiating antipsychotic treatment and recurrently for patients on

long-term antipsychotic therapy.

5.9 Leukopenia, Neutropenia, and Agranulocytosis

Class Effect: In clinical trial and/or postmarketing experience, events of leukopenia/neutropenia have been reported temporally related to antipsychotic agents, including RISPERDAL. Agranulocytosis has also been reported.

Possible risk factors for leukopenia/neutropenia include pre-existing low white blood cell count (WBC) and history of drug-induced leukopenia/neutropenia. Patients with a history of a clinically significant low WBC or a drug-induced leukopenia/neutropenia should have their complete blood count (CBC) monitored frequently during the first few months of therapy and discontinuation of RISPERDAL should be considered at the first sign of a clinically significant decline in WBC in the absence of other causative factors.

Patients with clinically significant neutropenia should be carefully monitored for fever or other symptoms or signs of infection and treated promptly if such symptoms or signs occur. Patients with severe neutropenia (absolute neutrophil count <1000/mm^3) should discontinue RISPERDAL and have their WBC followed until recovery.

5.10 Potential for Cognitive and Motor Impairment

Somnolence was a commonly reported adverse reaction associated with RISPERDAL treatment, especially when ascertained by direct questioning of patients. This adverse reaction is dose-related, and in a study utilizing a checklist to detect adverse events, 41% of the high-dose patients (RISPERDAL 16 mg/day) reported somnolence compared to 16% of placebo patients. Direct questioning is more sensitive for detecting adverse events than spontaneous reporting, by which 8% of RISPERDAL 16 mg/day patients and 1% of placebo patients reported somnolence as an adverse reaction. Since RISPERDAL has the potential to impair judgment, thinking, or motor skills, patients should be cautioned about operating hazardous machinery, including automobiles, until they are reasonably certain that RISPERDAL therapy does not affect them adversely.

5.11 Seizures

During premarketing testing in adult patients with schizophrenia, seizures occurred in 0.3% (9/2607) of RISPERDAL-treated patients, two in association with hyponatremia. RISPERDAL should be used cautiously in patients with a history of seizures.

5.12 Dysphagia

Esophageal dysmotility and aspiration have been associated with antipsychotic drug use. Aspiration pneumonia is a common cause of morbidity and mortality in patients with advanced Alzheimer's dementia. RISPERDAL and other antipsychotic drugs should be used cautiously in patients at risk for aspiration pneumonia. *[see Boxed Warning and Warnings and Precautions (5.1)]*.

5.13 Priapism

Priapism has been reported during postmarketing surveillance. Severe priapism may require surgical intervention.

5.14 Body Temperature Regulation

Disruption of body temperature regulation has been attributed to antipsychotic agents. Both hyperthermia and hypothermia have been reported in association with oral RISPERDAL use. Caution is advised when prescribing for patients who will be exposed to temperature extremes.

5.15 Patients with Phenylketonuria

Inform patients that RISPERDAL M-TAB Orally Disintegrating Tablets contain phenylalanine. Phenylalanine is a component of aspartame. Each 4 mg RISPERDAL M-TAB Orally Disintegrating Tablet contains 0.84 mg phenylalanine; each 3 mg RISPERDAL M-TAB Orally Disintegrating Tablet contains 0.63 mg phenylalanine; each 2 mg RISPERDAL M-TAB Orally Disintegrating Tablet contains 0.42 mg phenylalanine; each 1 mg RISPERDAL M-TAB Orally Disintegrating Tablet contains 0.28 mg phenylalanine; and each 0.5 mg RISPERDAL M-TAB Orally Disintegrating Tablet contains 0.14 mg phenylalanine.

6 ADVERSE REACTIONS

The following are discussed in more detail in other sections of the labeling:

- Increased mortality in elderly patients with dementia-related psychosis *[see Boxed Warning and Warnings and Precautions (5.1)]*
- Cerebrovascular adverse events, including stroke, in elderly patients with dementia-related psychosis *[see Warnings and Precautions (5.2)]*
- Neuroleptic malignant syndrome *[see Warnings and Precautions (5.3)]*
- Tardive dyskinesia *[see Warnings and Precautions (5.4)]*
- Metabolic Changes (Hyperglycemia and diabetes mellitus, Dyslipidemia, and Weight Gain) *[see Warnings and Precautions (5.5)]*
- Hyperprolactinemia *[see Warnings and Precautions (5.6)]*
- Orthostatic hypotension *[see Warnings and Precautions (5.7)]*
- Falls *[see Warnings and Precautions (5.8)]*
- Leukopenia, neutropenia, and agranulocytosis *[see Warnings and Precautions (5.9)]*
- Potential for cognitive and motor impairment *[see Warnings and Precautions (5.10)]*
- Seizures *[see Warnings and Precautions (5.11)]*
- Dysphagia *[see Warnings and Precautions (5.12)]*
- Priapism *[see Warnings and Precautions (5.13)]*
- Disruption of body temperature regulation *[see Warnings and Precautions (5.14)]*
- Patients with Phenylketonuria *[see Warnings and Precautions (5.15)].*

The most common adverse reactions in clinical trials (>5% and twice placebo) were parkinsonism, akathisia, dystonia, tremor, sedation, dizziness, anxiety, blurred vision, nausea, vomiting, upper abdominal pain, stomach discomfort, dyspepsia, diarrhea, salivary hypersecretion, constipation, dry mouth, increased appetite, increased weight, fatigue, rash, nasal congestion, upper respiratory tract infection, nasopharyngitis, and pharyngolaryngeal pain.

The most common adverse reactions that were associated with discontinuation from clinical trials (causing discontinuation in >1% of adults and/or >2% of pediatrics) were nausea, somnolence, sedation, vomiting, dizziness, and akathisia *[see Adverse Reactions, Discontinuations Due to Adverse Reactions (6.1)]*.

The data described in this section are derived from a clinical trial database consisting of 9803 adult and pediatric patients exposed to one or more doses of RISPERDAL for the treatment of schizophrenia, bipolar mania, autistic disorder, and other psychiatric disorders in pediatrics and elderly patients with dementia. Of these 9803 patients, 2687 were patients who received RISPERDAL while participating in double-blind, placebo-controlled trials. The conditions and duration of treatment with RISPERDAL varied greatly and included (in overlapping categories) double-blind, fixed- and flexible-dose, placebo- or active-controlled studies and open-label phases of studies, inpatients and outpatients, and short-term (up to 12 weeks) and longer-term (up to 3 years) exposures. Safety was assessed by collecting adverse events and performing physical examinations, vital signs, body weights, laboratory analyses, and ECGs.

6.1 Clinical Trials Experience

Because clinical trials are conducted under widely varying conditions, adverse reaction rates observed in the clinical trials of a drug cannot be directly compared to rates in the clinical trials of another drug and may not reflect the rates observed in clinical practice.

Commonly-Observed Adverse Reactions in Double-Blind, Placebo-Controlled Clinical Trials – Schizophrenia

Adult Patients with Schizophrenia

Table 8 lists the adverse reactions reported in 2% or more of RISPERDAL-treated adult patients with schizophrenia in three 4- to 8-week, double-blind, placebo-controlled trials.

Table 8. Adverse Reactions in ≥2% of RISPERDAL-Treated Adult Patients (and greater than placebo) with Schizophrenia in Double-Blind, Placebo-Controlled Trials

	Percentage of Patients Reporting Reaction		
	RISPERDAL		
System/Organ Class Adverse Reaction	**2–8 mg per day (N=366)**	**>8–16 mg per day (N=198)**	**Placebo (N=225)**
Cardiac Disorders			
Tachycardia	1	3	0
Eye Disorders			
Vision blurred	3	1	1
Gastrointestinal Disorders			
Nausea	9	4	4
Constipation	8	9	6
Dyspepsia	8	6	5
Dry mouth	4	0	1
Abdominal discomfort	3	1	1
Salivary hypersecretion	2	1	<1
Diarrhea	2	1	1
General Disorders			
Fatigue	3	1	0
Chest pain	2	2	1

Asthenia	2	1	<1
Infections and Infestations			
Nasopharyngitis	3	4	3
Upper respiratory tract infection	2	3	1
Sinusitis	1	2	1
Urinary tract infection	1	3	0
Investigations			
Blood creatine phosphokinase increased	1	2	<1
Heart rate increased	<1	2	0
Musculoskeletal and Connective Tissue Disorders			
Back pain	4	1	1
Arthralgia	2	3	<1
Pain in extremity	2	1	1
Nervous System Disorders			
Parkinsonism*	14	17	8
Akathisia*	10	10	3
Sedation	10	5	2
Dizziness	7	4	2
Dystonia*	3	4	2
Tremor*	2	3	1
Dizziness postural	2	0	0
Psychiatric Disorders			
Insomnia	32	25	27
Anxiety	16	11	11
Respiratory, Thoracic and Mediastinal Disorders			
Nasal congestion	4	6	2
Dyspnea	1	2	0
Epistaxis	<1	2	0
Skin and Subcutaneous Tissue Disorders			
Rash	1	4	1
Dry skin	1	3	0
Vascular Disorders			
Orthostatic hypotension	2	1	0

* Parkinsonism includes extrapyramidal disorder, musculoskeletal stiffness, parkinsonism, cogwheel rigidity, akinesia, bradykinesia, hypokinesia, masked facies, muscle rigidity, and Parkinson's disease. Akathisia includes akathisia and restlessness. Dystonia includes dystonia, muscle spasms, muscle contractions involuntary, muscle contracture, oculogyration, tongue paralysis. Tremor includes tremor and parkinsonian rest tremor.

Pediatric Patients with Schizophrenia

Table 9 lists the adverse reactions reported in 5% or more of RISPERDAL-treated pediatric patients with schizophrenia in a 6-week double-blind, placebo-controlled trial.

Table 9. Adverse Reactions in ≥5% of RISPERDAL-Treated Pediatric Patients (and greater than placebo) with Schizophrenia in a Double-Blind Trial

	Percentage of Patients Reporting Reaction		
	RISPERDAL		
System/Organ Class Adverse Reaction	1–3 mg per day (N=55)	4–6 mg per day (N=51)	Placebo (N=54)
Gastrointestinal Disorders			
Salivary hypersecretion	0	10	2
Nervous System Disorders			
Sedation	24	12	4
Parkinsonism*	16	28	11
Tremor	11	10	6
Akathisia*	9	10	4
Dizziness	7	14	2
Dystonia*	2	6	0
Psychiatric Disorders			
Anxiety	7	6	0

* Parkinsonism includes extrapyramidal disorder, muscle rigidity, musculoskeletal stiffness, and hypokinesia. Akathisia includes akathisia and restlessness. Dystonia includes dystonia and oculogyration.

Commonly-Observed Adverse Reactions in Double-Blind, Placebo-Controlled Clinical Trials – Bipolar Mania

Adult Patients with Bipolar Mania

Table 10 lists the adverse reactions reported in 2% or more of RISPERDAL-treated adult patients with bipolar mania in four 3-week, double-blind, placebo-controlled monotherapy trials.

Table 10. Adverse Reactions in ≥2% of RISPERDAL-Treated Adult Patients (and greater than placebo) with Bipolar Mania in Double-Blind, Placebo-Controlled Monotherapy Trials

	Percentage of Patients Reporting Reaction	
System/Organ Class Adverse Reaction	RISPERDAL 1–6 mg per day (N=448)	Placebo (N=424)
Eye Disorders		
Vision blurred	2	1
Gastrointestinal Disorders		
Nausea	5	2
Diarrhea	3	2
Salivary hypersecretion	3	1
Stomach discomfort	2	<1
General Disorders		

Fatigue	2	1
Nervous System Disorders		
Parkinsonism*	25	9
Sedation	11	4
Akathisia*	9	3
Tremor*	6	3
Dizziness	6	5
Dystonia*	5	1
Lethargy	2	1

* Parkinsonism includes extrapyramidal disorder, parkinsonism, musculoskeletal stiffness, hypokinesia, muscle rigidity, muscle tightness, bradykinesia, cogwheel rigidity. Akathisia includes akathisia and restlessness. Tremor includes tremor and parkinsonian rest tremor. Dystonia includes dystonia, muscle spasms, oculogyration, torticollis.

Table 11 lists the adverse reactions reported in 2% or more of RISPERDAL-treated adult patients with bipolar mania in two 3-week, double-blind, placebo-controlled adjuvant therapy trials.

Table 11. Adverse Reactions in ≥2% of RISPERDAL-Treated Adult Patients (and greater than placebo) with Bipolar Mania in Double-Blind, Placebo-Controlled Adjunctive Therapy Trials

	Percentage of Patients Reporting Reaction	
System/Organ Class Adverse Reaction	**RISPERDAL + Mood Stabilizer (N=127)**	**Placebo + Mood Stabilizer (N=126)**
Cardiac Disorders		
Palpitations	2	0
Gastrointestinal Disorders		
Dyspepsia	9	8
Nausea	6	4
Diarrhea	6	4
Salivary hypersecretion	2	0
General Disorders		
Chest pain	2	1
Infections and Infestations		
Urinary tract infection	2	1
Nervous System Disorders		
Parkinsonism*	14	4
Sedation	9	4
Akathisia*	8	0
Dizziness	7	2
Tremor	6	2
Lethargy	2	1
Psychiatric Disorders		
Anxiety	3	2
Respiratory, Thoracic and		

Mediastinal Disorders		
Pharyngolaryngeal pain	5	2
Cough	2	0

* Parkinsonism includes extrapyramidal disorder, hypokinesia and bradykinesia. Akathisia includes hyperkinesia and akathisia.

Pediatric Patients with Bipolar Mania

Table 12 lists the adverse reactions reported in 5% or more of RISPERDAL-treated pediatric patients with bipolar mania in a 3-week double-blind, placebo-controlled trial.

Table 12. Adverse Reactions in ≥5% of RISPERDAL-Treated Pediatric Patients (and greater than placebo) with Bipolar Mania in Double-Blind, Placebo-Controlled Trials

	Percentage of Patients Reporting Reaction		
	RISPERDAL		
System/Organ Class Adverse Reaction	**0.5–2.5 mg per day (N=50)**	**3–6 mg per day (N=61)**	**Placebo (N=58)**
Eye Disorders			
Vision blurred	4	7	0
Gastrointestinal Disorders			
Abdominal pain upper	16	13	5
Nausea	16	13	7
Vomiting	10	10	5
Diarrhea	8	7	2
Dyspepsia	10	3	2
Stomach discomfort	6	0	2
General Disorders			
Fatigue	18	30	3
Metabolism and Nutrition Disorders			
Increased appetite	4	7	2
Nervous System Disorders			
Sedation	42	56	19
Dizziness	16	13	5
Parkinsonism*	6	12	3
Dystonia*	6	5	0
Akathisia*	0	8	2
Psychiatric Disorders			
Anxiety	0	8	3
Respiratory, Thoracic and Mediastinal Disorders			
Pharyngolaryngeal pain	10	3	5
Skin and Subcutaneous Tissue Disorders			

Rash	0	7	2

* Parkinsonism includes musculoskeletal stiffness, extrapyramidal disorder, bradykinesia, and nuchal rigidity. Dystonia includes dystonia, laryngospasm, and muscle spasms. Akathisia includes restlessness and akathisia.

Commonly-Observed Adverse Reactions in Double-Blind, Placebo-Controlled Clinical Trials - Autistic Disorder

Table 13 lists the adverse reactions reported in 5% or more of RISPERDAL-treated pediatric patients treated for irritability associated with autistic disorder in two 8-week, double-blind, placebo-controlled trials and one 6-week double-blind, placebo-controlled study.

Table 13. Adverse Reactions in ≥5% of RISPERDAL-Treated Pediatric Patients (and greater than placebo) Treated for Irritability Associated with Autistic Disorder in Double-Blind, Placebo-Controlled Trials

	Percentage of Patients Reporting Reaction	
System/Organ Class Adverse Reaction	**RISPERDAL 0.5–4.0 mg/day (N=107)**	**Placebo (N=115)**
Gastrointestinal Disorders		
Vomiting	20	17
Constipation	17	6
Dry mouth	10	4
Nausea	8	5
Salivary hypersecretion	7	1
General Disorders and Administration Site Conditions		
Fatigue	31	9
Pyrexia	16	13
Thirst	7	4
Infections and Infestations		
Nasopharyngitis	19	9
Rhinitis	9	7
Upper respiratory tract infection	8	3
Investigations		
Weight increased	8	2
Metabolism and Nutrition Disorders		
Increased appetite	44	15
Nervous System Disorders		
Sedation	63	15
Drooling	12	4
Headache	12	10
Tremor	8	1
Dizziness	8	2
Parkinsonism*	8	1

Renal and Urinary Disorders		
Enuresis	16	10
Respiratory, Thoracic and Mediastinal Disorders		
Cough	17	12
Rhinorrhea	12	10
Nasal congestion	10	4
Skin and Subcutaneous Tissue Disorders		
Rash	8	5

* Parkinsonism includes musculoskeletal stiffness, extrapyramidal disorder, muscle rigidity, cogwheel rigidity, and muscle tightness.

Other Adverse Reactions Observed During the Clinical Trial Evaluation of Risperidone

The following additional adverse reactions occurred across all placebo-controlled, active-controlled, and open-label studies of RISPERDAL in adults and pediatric patients.

Blood and Lymphatic System Disorders: anemia, granulocytopenia, neutropenia

Cardiac Disorders: sinus bradycardia, sinus tachycardia, atrioventricular block first degree, bundle branch block left, bundle branch block right, atrioventricular block

Ear and Labyrinth Disorders: ear pain, tinnitus

Endocrine Disorders: hyperprolactinemia

Eye Disorders: ocular hyperemia, eye discharge, conjunctivitis, eye rolling, eyelid edema, eye swelling, eyelid margin crusting, dry eye, lacrimation increased, photophobia, glaucoma, visual acuity reduced

Gastrointestinal Disorders: dysphagia, fecaloma, fecal incontinence, gastritis, lip swelling, cheilitis, aptyalism

General Disorders: edema peripheral, thirst, gait disturbance, influenza-like illness, pitting edema, edema, chills, sluggishness, malaise, chest discomfort, face edema, discomfort, generalized edema, drug withdrawal syndrome, peripheral coldness, feeling abnormal

Immune System Disorders: drug hypersensitivity

Infections and Infestations: pneumonia, influenza, ear infection, viral infection, pharyngitis, tonsillitis, bronchitis, eye infection, localized infection, cystitis, cellulitis, otitis media, onychomycosis, acarodermatitis, bronchopneumonia, respiratory tract infection, tracheobronchitis, otitis media chronic

Investigations: body temperature increased, blood prolactin increased, alanine aminotransferase increased, electrocardiogram abnormal, eosinophil count increased, white blood cell count decreased, blood glucose increased, hemoglobin decreased, hematocrit decreased, body temperature decreased, blood pressure decreased, transaminases increased

Metabolism and Nutrition Disorders: decreased appetite, polydipsia, anorexia

Musculoskeletal and Connective Tissue Disorders: joint stiffness, joint swelling, musculoskeletal chest pain, posture abnormal, myalgia, neck pain, muscular weakness, rhabdomyolysis

Nervous System Disorders: balance disorder, disturbance in attention, dysarthria, unresponsive to stimuli, depressed level of consciousness, movement disorder, transient ischemic attack, coordination abnormal, cerebrovascular accident, speech disorder, syncope, loss of consciousness, hypoesthesia, tardive dyskinesia, dyskinesia, cerebral ischemia, cerebrovascular disorder, neuroleptic malignant syndrome, diabetic coma, head titubation

Psychiatric Disorders: agitation, blunted affect, confusional state, middle insomnia, nervousness, sleep disorder, listlessness, libido decreased, and anorgasmia

Renal and Urinary Disorders: enuresis, dysuria, pollakiuria, urinary incontinence

Reproductive System and Breast Disorders: menstruation irregular, amenorrhea, gynecomastia, galactorrhea, vaginal discharge, menstrual disorder, erectile dysfunction, retrograde ejaculation, ejaculation disorder, sexual dysfunction, breast enlargement

Respiratory, Thoracic, and Mediastinal Disorders: wheezing, pneumonia aspiration, sinus congestion, dysphonia, productive cough, pulmonary congestion, respiratory tract congestion, rales, respiratory disorder, hyperventilation, nasal edema

Skin and Subcutaneous Tissue Disorders: erythema, skin discoloration, skin lesion, pruritus, skin disorder, rash erythematous, rash papular, rash generalized, rash maculopapular, acne, hyperkeratosis, seborrheic dermatitis

Vascular Disorders: hypotension, flushing

Additional Adverse Reactions Reported with RISPERDAL CONSTA

The following is a list of additional adverse reactions that have been reported during the premarketing evaluation of RISPERDAL CONSTA, regardless of frequency of occurrence:

Cardiac Disorders: bradycardia

Ear and Labyrinth Disorders: vertigo

Eye Disorders: blepharospasm

Gastrointestinal Disorders: toothache, tongue spasm

General Disorders and Administration Site Conditions: pain

Infections and Infestations: lower respiratory tract infection, infection, gastroenteritis, subcutaneous abscess

Injury and Poisoning: fall

Investigations: weight decreased, gamma-glutamyltransferase increased, hepatic enzyme increased

Musculoskeletal, Connective Tissue, and Bone Disorders: buttock pain

Nervous System Disorders: convulsion, paresthesia

Psychiatric Disorders: depression

Skin and Subcutaneous Tissue Disorders: eczema

Vascular Disorders: hypertension

Discontinuations Due to Adverse Reactions

Schizophrenia - Adults

Approximately 7% (39/564) of RISPERDAL-treated patients in double-blind, placebo-controlled trials discontinued treatment due to an adverse reaction, compared with 4% (10/225) who were receiving placebo. The adverse reactions associated with discontinuation in 2 or more RISPERDAL-treated patients were:

Table 14. Adverse Reactions Associated With Discontinuation in 2 or More RISPERDAL-Treated Adult Patients in Schizophrenia Trials

	RISPERDAL		
Adverse Reaction	**2–8 mg/day (N=366)**	**>8–16 mg/day (N=198)**	**Placebo (N=225)**
Dizziness	1.4%	1.0%	0%
Nausea	1.4%	0%	0%
Vomiting	0.8%	0%	0%
Parkinsonism	0.8%	0%	0%
Somnolence	0.8%	0%	0%
Dystonia	0.5%	0%	0%
Agitation	0.5%	0%	0%
Abdominal pain	0.5%	0%	0%
Orthostatic hypotension	0.3%	0.5%	0%
Akathisia	0.3%	2.0%	0%

Discontinuation for extrapyramidal symptoms (including Parkinsonism, akathisia, dystonia, and tardive dyskinesia) was 1% in placebo-treated patients, and 3.4% in active control-treated patients in a double-blind, placebo- and active-controlled trial.

Schizophrenia - Pediatrics

Approximately 7% (7/106), of RISPERDAL-treated patients discontinued treatment due to an adverse reaction in a double-blind, placebo-controlled trial, compared with 4% (2/54) placebo-treated patients. The adverse reactions associated with discontinuation for at least one RISPERDAL-treated patient were dizziness (2%), somnolence (1%), sedation (1%), lethargy (1%), anxiety (1%), balance disorder (1%), hypotension (1%), and palpitation (1%).

Bipolar Mania - Adults

In double-blind, placebo-controlled trials with RISPERDAL as monotherapy, approximately 6% (25/448) of RISPERDAL-treated patients discontinued treatment due to an adverse event, compared with approximately 5% (19/424) of placebo-treated patients. The adverse reactions associated with discontinuation in RISPERDAL-treated patients were:

Table 15. Adverse Reactions Associated With Discontinuation in 2 or More RISPERDAL-Treated Adult Patients in Bipolar Mania Clinical Trials

	RISPERDAL

Adverse Reaction	1–6 mg/day (N=448)	Placebo (N=424)
Parkinsonism	0.4%	0%
Lethargy	0.2%	0%
Dizziness	0.2%	0%
Alanine aminotransferase increased	0.2%	0.2%
Aspartate aminotransferase increased	0.2%	0.2%

Bipolar Mania - Pediatrics

In a double-blind, placebo-controlled trial 12% (13/111) of RISPERDAL-treated patients discontinued due to an adverse reaction, compared with 7% (4/58) of placebo-treated patients. The adverse reactions associated with discontinuation in more than one RISPERDAL-treated pediatric patient were nausea (3%), somnolence (2%), sedation (2%), and vomiting (2%).

Autistic Disorder - Pediatrics

In the two 8-week, placebo-controlled trials in pediatric patients treated for irritability associated with autistic disorder (n=156), one RISPERDAL-treated patient discontinued due to an adverse reaction (Parkinsonism), and one placebo-treated patient discontinued due to an adverse event.

Dose Dependency of Adverse Reactions in Clinical Trials

Extrapyramidal Symptoms

Data from two fixed-dose trials in adults with schizophrenia provided evidence of dose-relatedness for extrapyramidal symptoms associated with RISPERDAL treatment.

Two methods were used to measure extrapyramidal symptoms (EPS) in an 8-week trial comparing 4 fixed doses of RISPERDAL (2, 6, 10, and 16 mg/day), including (1) a Parkinsonism score (mean change from baseline) from the Extrapyramidal Symptom Rating Scale, and (2) incidence of spontaneous complaints of EPS:

Table 16.

Dose Groups	Placebo	RISPERDAL 2 mg	RISPERDAL 6 mg	RISPERDAL 10 mg	RISPERDAL 16 mg
Parkinsonism	1.2	0.9	1.8	2.4	2.6
EPS Incidence	13%	17%	21%	21%	35%

Similar methods were used to measure extrapyramidal symptoms (EPS) in an 8-week trial comparing 5 fixed doses of RISPERDAL (1, 4, 8, 12, and 16 mg/day):

Table 17.

Dose Groups	RISPERDAL 1 mg	RISPERDAL 4 mg	RISPERDAL 8 mg	RISPERDAL 12 mg	RISPERDAL 16 mg
Parkinsonism	0.6	1.7	2.4	2.9	4.1

EPS Incidence	7%	12%	17%	18%	20%

Dystonia

Class Effect: Symptoms of dystonia, prolonged abnormal contractions of muscle groups, may occur in susceptible individuals during the first few days of treatment. Dystonic symptoms include: spasm of the neck muscles, sometimes progressing to tightness of the throat, swallowing difficulty, difficulty breathing, and/or protrusion of the tongue. While these symptoms can occur at low doses, they occur more frequently and with greater severity with high potency and at higher doses of first generation antipsychotic drugs. An elevated risk of acute dystonia is observed in males and younger age groups.

Other Adverse Reactions

Adverse event data elicited by a checklist for side effects from a large study comparing 5 fixed doses of RISPERDAL (1, 4, 8, 12, and 16 mg/day) were explored for dose-relatedness of adverse events. A Cochran-Armitage Test for trend in these data revealed a positive trend ($p<0.05$) for the following adverse reactions: somnolence, vision abnormal, dizziness, palpitations, weight increase, erectile dysfunction, ejaculation disorder, sexual function abnormal, fatigue, and skin discoloration.

Changes in Body Weight

Weight gain was observed in short-term, controlled trials and longer-term uncontrolled studies in adult and pediatric patients *[see Warnings and Precautions (5.5), Adverse Reactions (6), and Use in Specific Populations (8.4)].*

Changes in ECG Parameters

Between-group comparisons for pooled placebo-controlled trials in adults revealed no statistically significant differences between risperidone and placebo in mean changes from baseline in ECG parameters, including QT, QTc, and PR intervals, and heart rate. When all RISPERDAL doses were pooled from randomized controlled trials in several indications, there was a mean increase in heart rate of 1 beat per minute compared to no change for placebo patients. In short-term schizophrenia trials, higher doses of risperidone (8–16 mg/day) were associated with a higher mean increase in heart rate compared to placebo (4–6 beats per minute). In pooled placebo-controlled acute mania trials in adults, there were small decreases in mean heart rate, similar among all treatment groups.

In the two placebo-controlled trials in children and adolescents with autistic disorder (aged 5 – 16 years) mean changes in heart rate were an increase of 8.4 beats per minute in the RISPERDAL groups and 6.5 beats per minute in the placebo group. There were no other notable ECG changes.

In a placebo-controlled acute mania trial in children and adolescents (aged 10 – 17 years), there were no significant changes in ECG parameters, other than the effect of RISPERDAL to transiently increase pulse rate (< 6 beats per minute). In two controlled schizophrenia trials in adolescents (aged 13 – 17 years), there were no clinically meaningful changes in ECG parameters including corrected QT intervals between treatment groups or within treatment groups over time.

6.2 Postmarketing Experience

The following adverse reactions have been identified during postapproval use of risperidone. Because these reactions are reported voluntarily from a population of uncertain size, it is not always possible to reliably estimate their frequency or establish a causal relationship to drug exposure. These adverse reactions include: alopecia, anaphylactic reaction, angioedema, atrial fibrillation, cardiopulmonary arrest, catatonia, diabetic ketoacidosis in patients with impaired glucose metabolism, dysgeusia, hypoglycemia, hypothermia, ileus, inappropriate antidiuretic hormone secretion, intestinal obstruction, jaundice, mania, pancreatitis, pituitary adenoma, precocious puberty, pulmonary embolism, QT prolongation, sleep apnea syndrome, somnambulism, Stevens-Johnson syndrome and toxic epidermal necrolysis (SJS/TEN), sudden death, thrombocytopenia, thrombotic thrombocytopenic purpura, urinary retention, and water intoxication.

Postmarketing cases of extrapyramidal symptoms (dystonia and dyskinesia) have been reported in patients concomitantly taking methylphenidate and risperidone when there was an increase or decrease in dosage, initiation, or discontinuation of either or both medications.

7 DRUG INTERACTIONS

7.1 Pharmacokinetic-related Interactions

The dose of RISPERDAL should be adjusted when used in combination with CYP2D6 enzyme inhibitors (e.g., fluoxetine, and paroxetine) and enzyme inducers (e.g., carbamazepine) *[see Table 18 and Dosage and Administration (2.5)]*. Dose adjustment is not recommended for RISPERDAL when co-administered with ranitidine, cimetidine, amitriptyline, or erythromycin *[see Table 18].*

Table 18. Summary of Effect of Coadministered Drugs on Exposure to Active Moiety (Risperidone + 9-Hydroxy-Risperidone) in Healthy Subjects or Patients with Schizophrenia

Coadministered Drug	**Dosing Schedule**		**Effect on Active Moiety (Risperidone + 9-Hydroxy-Risperidone (Ratio*))**		**Risperidone Dose Recommendation**
	Coadministered Drug	**Risperidone**	**AUC**	**C_{max}**	
Enzyme (CYP2D6) Inhibitors					
Fluoxetine	20 mg/day	2 or 3 mg twice daily	1.4	1.5	Re-evaluate dosing. Do not exceed 8 mg/day
Paroxetine	10 mg/day	4 mg/day	1.3	-	Re-evaluate dosing. Do not exceed 8 mg/day
	20 mg/day	4 mg/day	1.6	-	
	40 mg/day	4 mg/day	1.8	-	

Enzyme (CYP3A/PgP inducers) Inducers					
Carbamazepine	573 ± 168 mg/day	3 mg twice daily	0.51	0.55	Titrate dose upwards. Do not exceed twice the patient's usual dose
Enzyme (CYP3A) Inhibitors					
Ranitidine	150 mg twice daily	1 mg single dose	1.2	1.4	Dose adjustment not needed
Cimetidine	400 mg twice daily	1 mg single dose	1.1	1.3	Dose adjustment not needed
Erythromycin	500 mg four times daily	1 mg single dose	1.1	0.94	Dose adjustment not needed
Other Drugs					
Amitriptyline	50 mg twice daily	3 mg twice daily	1.2	1.1	Dose adjustment not needed

* Change relative to reference

Effect of Risperidone on Other Drugs

Lithium

Repeated oral doses of RISPERDAL (3 mg twice daily) did not affect the exposure (AUC) or peak plasma concentrations (C_{max}) of lithium (n=13). Dose adjustment for lithium is not recommended.

Valproate

Repeated oral doses of RISPERDAL (4 mg once daily) did not affect the pre-dose or average plasma concentrations and exposure (AUC) of valproate (1000 mg/day in three divided doses) compared to placebo (n=21). However, there was a 20% increase in valproate peak plasma concentration (C_{max}) after concomitant administration of RISPERDAL. Dose adjustment for valproate is not recommended.

Digoxin

RISPERDAL (0.25 mg twice daily) did not show a clinically relevant effect on the pharmacokinetics of digoxin. Dose adjustment for digoxin is not recommended.

7.2 Pharmacodynamic-related Interactions

Centrally Acting Drugs and Alcohol

Given the primary CNS effects of risperidone, caution should be used when RISPERDAL is taken in combination with other centrally acting drugs and alcohol.

Drugs with Hypotensive Effects

Because of its potential for inducing hypotension, RISPERDAL may enhance the hypotensive effects of other therapeutic agents with this potential.

Levodopa and Dopamine Agonists

RISPERDAL may antagonize the effects of levodopa and dopamine agonists.

Methylphenidate

Concomitant use with methylphenidate, when there is change in dosage of either medication, may increase the risk of extrapyramidal symptoms (EPS). Monitor for symptoms of EPS with concomitant use of RISPERDAL and methylphenidate *[see Adverse Reactions (6.2)]*.

Clozapine

Chronic administration of clozapine with RISPERDAL may decrease the clearance of risperidone.

8 USE IN SPECIFIC POPULATIONS

8.1 Pregnancy

Pregnancy Exposure Registry

There is a pregnancy exposure registry that monitors pregnancy outcomes in women exposed to atypical antipsychotics, including RISPERDAL, during pregnancy. Healthcare providers are encouraged to register patients by contacting the National Pregnancy Registry for Atypical Antipsychotics at 1-866-961-2388 or online at *http://womensmentalhealth.org/clinical-and-research-programs/pregnancyregistry/*.

Risk Summary

Neonates exposed to antipsychotic drugs during the third trimester of pregnancy are at risk for extrapyramidal and/or withdrawal symptoms following delivery *(see Clinical Considerations)*. Overall, available data from published epidemiologic studies of pregnant women exposed to risperidone have not established a drug-associated risk of major birth defects, miscarriage, or adverse maternal or fetal outcomes *(see Data).* There are risks to the mother associated with untreated schizophrenia or bipolar I disorder and with exposure to antipsychotics, including RISPERDAL, during pregnancy *(see Clinical Considerations)*.

Oral administration of risperidone to pregnant mice caused cleft palate at doses 3 to 4 times the maximum recommended human dose (MRHD) with maternal toxicity observed at 4-times MRHD based on mg/m^2 body surface area. Risperidone was not teratogenic in rats or rabbits at doses up to 6-times the MRHD based on mg/m^2 body surface area. Increased stillbirths and decreased birth weight occurred after oral risperidone administration to pregnant rats at 1.5-times the MRHD based on mg/m^2 body surface area. Learning was impaired in offspring of rats when the dams were dosed at 0.6-times the MRHD and offspring mortality increased at doses 0.1 to 3 times the MRHD based on mg/m^2 body surface area.

The estimated background risk of major birth defects and miscarriage for the indicated population is unknown. All pregnancies have a background risk of birth defect, loss, or other adverse outcomes. In the U.S. general population, the estimated background risk of major birth defects and miscarriage in clinically recognized pregnancies is 2–4% and 15–20%, respectively.

Clinical Considerations

Disease-associated maternal and/or embryo/fetal risk

There is a risk to the mother from untreated schizophrenia or bipolar I disorder, including increased risk of relapse, hospitalization, and suicide. Schizophrenia and bipolar I disorder are associated with increased adverse perinatal outcomes, including preterm birth. It is not known if this is a direct result of the illness or other comorbid factors.

Fetal/Neonatal Adverse Reactions

Extrapyramidal and/or withdrawal symptoms, including agitation, hypertonia, hypotonia, tremor, somnolence, respiratory distress, and feeding disorder have been reported in neonates who were exposed to antipsychotic drugs, including RISPERDAL, during the third trimester of pregnancy. These symptoms have varied in severity. Monitor neonates for extrapyramidal and/or withdrawal symptoms and manage symptoms appropriately. Some neonates recovered within hours or days without specific treatment; others required prolonged hospitalization.

Data

Human Data

Published data from observational studies, birth registries, and case reports on the use of atypical antipsychotics during pregnancy do not report a clear association with antipsychotics and major birth defects. A prospective observational study including 6 women treated with risperidone demonstrated placental passage of risperidone. A retrospective cohort study from a Medicaid database of 9258 women exposed to antipsychotics during pregnancy did not indicate an overall increased risk for major birth defects. There was a small increase in the risk of major birth defects (RR=1.26, 95% CI 1.02–1.56) and of cardiac malformations (RR=1.26, 95% CI 0.88–1.81) in a subgroup of 1566 women exposed to risperidone during the first trimester of pregnancy; however, there is no mechanism of action to explain the difference in malformation rates.

Animal Data

Oral administration of risperidone to pregnant mice during organogenesis caused cleft palate at 10 mg/kg/day which is 3 times the MRHD of 16 mg/day based on mg/m^2 body surface area: maternal toxicity occurred at 4 times the MRHD. Risperidone was not teratogenic when administered orally to rats at 0.6 to 10 mg/kg/day and rabbits at 0.3 to 5 mg/kg/day, which are up to 6 times the MRHD of 16 mg/day risperidone based on mg/m^2 body surface area. Learning was impaired in offspring of rats dosed orally throughout pregnancy at 1 mg/kg/day which is 0.6 times the MRHD and neuronal cell death increased in fetal brains of offspring of rats dosed during pregnancy at 1 and 2 mg/kg/day which are 0.6 and 1.2 times the MRHD based on mg/m^2 body surface area; postnatal development and growth of the offspring were also delayed.

Rat offspring mortality increased during the first 4 days of lactation when pregnant rats were dosed throughout gestation at 0.16 to 5 mg/kg/day which are 0.1 to 3 times the MRHD of 16 mg/day based on mg/m^2 body surface area. It is not known whether these deaths were due to a direct effect on the fetuses or pups or to effects on the dams; a no-effect dose could not be determined. The rate of stillbirths was increased at 2.5 mg/kg or 1.5 times the MRHD based on mg/m^2 body surface area.

In a rat cross-fostering study the number of live offspring was decreased, the number

of stillbirths increased, and the birth weight was decreased in offspring of drug-treated pregnant rats. In addition, the number of deaths increased by Day 1 among offspring of drug-treated pregnant rats, regardless of whether or not the offspring were cross-fostered. Risperidone also appeared to impair maternal behavior in that offspring body weight gain and survival (from Day 1 to 4 of lactation) were reduced in offspring born to control but reared by drug-treated dams. All of these effects occurred at 5 mg/kg which is 3 times the MRHD based on mg/m^2 and the only dose tested in the study.

8.2 Lactation

Risk Summary

Limited data from published literature reports the presence of risperidone and its metabolite, 9-hydroxyrisperidone, in human breast milk at relative infant dose ranging between 2.3% and 4.7% of the maternal weight-adjusted dosage. There are reports of sedation, failure to thrive, jitteriness, and extrapyramidal symptoms (tremors and abnormal muscle movements) in breastfed infants exposed to risperidone *(see Clinical Considerations).* There is no information on the effects of risperidone on milk production. The developmental and health benefits of breastfeeding should be considered along with the mother's clinical need for RISPERDAL and any potential adverse effects on the breastfed child from RISPERDAL or from the mother's underlying condition.

Clinical Considerations

Infants exposed to RISPERDAL through breastmilk should be monitored for excess sedation, failure to thrive, jitteriness, and extrapyramidal symptoms (tremors and abnormal muscle movements).

8.3 Females and Males of Reproductive Potential

Infertility

Females

Based on the pharmacologic action of risperidone (D_2 receptor antagonism), treatment with RISPERDAL may result in an increase in serum prolactin levels, which may lead to a reversible reduction in fertility in females of reproductive potential *[see Warnings and Precautions (5.6)].*

8.4 Pediatric Use

Approved Pediatric Indications

Schizophrenia

The efficacy and safety of RISPERDAL in the treatment of schizophrenia were demonstrated in 417 adolescents, aged 13 to 17 years, in two short-term (6 and 8 weeks, respectively) double-blind controlled trials *[see Indications and Usage (1.1)*, *Adverse Reactions (6.1)*, *and Clinical Studies (14.1)].* Additional safety and efficacy information was also assessed in one long-term (6-month) open-label extension study in 284 of these adolescent patients with schizophrenia.

Safety and effectiveness of RISPERDAL in children less than 13 years of age with schizophrenia have not been established.

Bipolar I Disorder

The efficacy and safety of RISPERDAL in the short-term treatment of acute manic or mixed episodes associated with Bipolar I Disorder in 169 children and adolescent patients, aged 10 to 17 years, were demonstrated in one double-blind, placebo-controlled, 3-week trial *[see Indications and Usage (1.2), Adverse Reactions (6.1), and Clinical Studies (14.2)]*.

Safety and effectiveness of RISPERDAL in children less than 10 years of age with bipolar disorder have not been established.

Autistic Disorder

The efficacy and safety of RISPERDAL in the treatment of irritability associated with autistic disorder were established in two 8-week, double-blind, placebo-controlled trials in 156 children and adolescent patients, aged 5 to 16 years *[see Indications and Usage (1.3), Adverse Reactions (6.1) and Clinical Studies (14.4)]*. Additional safety information was also assessed in a long-term study in patients with autistic disorder, or in short- and long-term studies in more than 1200 pediatric patients with psychiatric disorders other than autistic disorder, schizophrenia, or bipolar mania who were of similar age and weight, and who received similar dosages of RISPERDAL as patients treated for irritability associated with autistic disorder.

A third study was a 6-week, multicenter, randomized, double-blind, placebo-controlled, fixed-dose study to evaluate the efficacy and safety of a lower than recommended dose of risperidone in subjects 5 to 17 years of age with autistic disorder and associated irritability, and related behavioral symptoms. There were two weight-based, fixed doses of risperidone (high-dose and low-dose). The high dose was 1.25 mg per day for patients weighing 20 to < 45 kg, and it was 1.75 mg per day for patients weighing ≥ 45 kg. The low dose was 0.125 mg per day for patients for patients weighing 20 to < 45 kg, and it was 0.175 mg per day for patients weighing ≥ 45 kg. The study demonstrated the efficacy of high-dose risperidone, but it did not demonstrate efficacy for low-dose risperidone.

Adverse Reactions in Pediatric Patients

Tardive Dyskinesia

In clinical trials in 1885 children and adolescents treated with RISPERDAL, 2 (0.1%) patients were reported to have tardive dyskinesia, which resolved on discontinuation of RISPERDAL treatment *[see also Warnings and Precautions (5.4)]*.

Weight Gain

Weight gain has been observed in children and adolescents during treatment with RISPERDAL. Clinical monitoring of weight is recommended during treatment.

Data derive from short-term placebo-controlled trials and longer-term uncontrolled studies in pediatric patients (ages 5 to 17 years) with schizophrenia, bipolar disorder, autistic disorder, or other psychiatric disorders. In the short-term trials (3 to 8 weeks), the mean weight gain for RISPERDAL-treated patients was 2 kg, compared to 0.6 kg for placebo-treated patients. In these trials, approximately 33% of the RISPERDAL group had weight gain ≥7%, compared to 7% in the placebo group. In longer-term, uncontrolled, open-label pediatric studies, the mean weight gain was 5.5 kg at Week 24 and 8 kg at Week 48 *[see Warnings and Precautions (5.5) and Adverse Reactions (6.1)].*

Somnolence

Somnolence was frequently observed in placebo-controlled clinical trials of pediatric patients with autistic disorder. Most cases were mild or moderate in severity. These events were most often of early onset with peak incidence occurring during the first two weeks of treatment, and transient with a median duration of 16 days. Somnolence was the most commonly observed adverse reaction in the clinical trial of bipolar disorder in children and adolescents, as well as in the schizophrenia trials in adolescents. As was seen in the autistic disorder trials, these adverse reactions were most often of early onset and transient in duration *[see Adverse Reactions (6.1 and 6.2)]*. Patients experiencing persistent somnolence may benefit from a change in dosing regimen *[see Dosage and Administration (2.1, 2.2, and 2.3)]*.

Hyperprolactinemia

RISPERDAL has been shown to elevate prolactin levels in children and adolescents as well as in adults *[see Warnings and Precautions (5.6)]*. In double-blind, placebo-controlled studies of up to 8 weeks duration in children and adolescents (aged 5 to 17 years) with autistic disorder or psychiatric disorders other than autistic disorder, schizophrenia, or bipolar mania, 49% of patients who received RISPERDAL had elevated prolactin levels compared to 2% of patients who received placebo. Similarly, in placebo-controlled trials in children and adolescents (aged 10 to 17 years) with bipolar disorder, or adolescents (aged 13 to 17 years) with schizophrenia, 82–87% of patients who received RISPERDAL had elevated levels of prolactin compared to 3–7% of patients on placebo. Increases were dose-dependent and generally greater in females than in males across indications.

In clinical trials in 1885 children and adolescents, galactorrhea was reported in 0.8% of RISPERDAL-treated patients and gynecomastia was reported in 2.3% of RISPERDAL-treated patients.

Growth and Sexual Maturation

The long-term effects of RISPERDAL on growth and sexual maturation have not been fully evaluated in children and adolescents.

Juvenile Animal Studies

Juvenile dogs were treated with oral risperidone from weeks 10 to 50 of age (equivalent to the period of childhood through adolescence in humans), at doses of 0.31, 1.25, or 5 mg/kg/day, which are 1.2, 3.4, and 13.5 times the MRHD of 6 mg/day for children, based on mg/m^2 body surface area. Bone length and density were decreased with a no-effect dose of 0.31 mg/kg/day; this dose produced plasma AUC of risperidone plus its active metabolite paliperidone (9-hydroxy-risperidone) that were similar to those in children and adolescents receiving the MRHD of 6 mg/day. In addition, sexual maturation was delayed at all doses in both males and females. The above effects showed little or no reversibility in females after a 12 week drug-free recovery period.

Juvenile rats, treated with oral risperidone from days 12 to 50 of age (equivalent to the period of infancy through adolescence in humans) showed impaired learning and memory performance (reversible only in females), with a no-effect dose of 0.63 mg/kg/day which is 0.5 times the MRHD of 6 mg/day for children, based on mg/m^2 body surface area. This dose produced plasma AUC of risperidone plus paliperidone about half the exposure observed in humans at the MRHD. No other consistent effects on neurobehavioral or reproductive development were seen up to the highest tested dose

of 1.25 mg/kg/day which is 1 time the MRHD and produced plasma AUC of risperidone plus paliperidone that were about two thirds of those observed in humans at the MRHD of 6 mg/day for children.

8.5 Geriatric Use

Clinical studies of RISPERDAL in the treatment of schizophrenia did not include sufficient numbers of patients aged 65 and over to determine whether or not they respond differently than younger patients. Other reported clinical experience has not identified differences in responses between elderly and younger patients. In general, a lower starting dose is recommended for an elderly patient, reflecting a decreased pharmacokinetic clearance in the elderly, as well as a greater frequency of decreased hepatic, renal, or cardiac function, and of concomitant disease or other drug therapy *[see Clinical Pharmacology (12.3) and Dosage and Administration (2.4, 2.5)]*. While elderly patients exhibit a greater tendency to orthostatic hypotension, its risk in the elderly may be minimized by limiting the initial dose to 0.5 mg twice daily followed by careful titration *[see Warnings and Precautions (5.7)]*. Monitoring of orthostatic vital signs should be considered in patients for whom this is of concern.

This drug is substantially excreted by the kidneys, and the risk of toxic reactions to this drug may be greater in patients with impaired renal function. Because elderly patients are more likely to have decreased renal function, care should be taken in dose selection, and it may be useful to monitor renal function *[see Dosage and Administration (2.4)]*.

8.6 Renal Impairment

In patients with moderate to severe (Clcr 59 to 15 mL/min) renal disease, clearance of the sum of risperidone and its active metabolite decreased by 60%, compared to young healthy subjects. RISPERDAL doses should be reduced in patients with renal disease *[see Dosage and Administration (2.4)]*.

8.7 Hepatic Impairment

While the pharmacokinetics of risperidone in subjects with liver disease were comparable to those in young healthy subjects, the mean free fraction of risperidone in plasma was increased by about 35% because of the diminished concentration of both albumin and α_1-acid glycoprotein. RISPERDAL doses should be reduced in patients with liver disease *[see Dosage and Administration (2.4)]*.

8.8 Patients with Parkinson's Disease or Lewy Body Dementia

Patients with Parkinson's Disease or Dementia with Lewy Bodies can experience increased sensitivity to RISPERDAL. Manifestations can include confusion, obtundation, postural instability with frequent falls, extrapyramidal symptoms, and clinical features consistent with neuroleptic malignant syndrome.

9 DRUG ABUSE AND DEPENDENCE

9.1 Controlled Substance

RISPERDAL (risperidone) is not a controlled substance.

9.2 Abuse

RISPERDAL has not been systematically studied in animals or humans for its potential for abuse. While the clinical trials did not reveal any tendency for any drug-seeking behavior, these observations were not systematic and it is not possible to predict on the basis of this limited experience the extent to which a CNS-active drug will be misused, diverted, and/or abused once marketed. Consequently, patients should be evaluated carefully for a history of drug abuse, and such patients should be observed closely for signs of RISPERDAL misuse or abuse (e.g., development of tolerance, increases in dose, drug-seeking behavior).

9.3 Dependence

RISPERDAL has not been systematically studied in animals or humans for its potential for tolerance or physical dependence.

10 OVERDOSAGE

10.1 Human Experience

Premarketing experience included eight reports of acute RISPERDAL overdosage with estimated doses ranging from 20 to 300 mg and no fatalities. In general, reported signs and symptoms were those resulting from an exaggeration of the drug's known pharmacological effects, i.e., drowsiness and sedation, tachycardia and hypotension, and extrapyramidal symptoms. One case, involving an estimated overdose of 240 mg, was associated with hyponatremia, hypokalemia, prolonged QT, and widened QRS. Another case, involving an estimated overdose of 36 mg, was associated with a seizure.

Postmarketing experience includes reports of acute RISPERDAL overdosage, with estimated doses of up to 360 mg. In general, the most frequently reported signs and symptoms are those resulting from an exaggeration of the drug's known pharmacological effects, i.e., drowsiness, sedation, tachycardia, hypotension, and extrapyramidal symptoms. Other adverse reactions reported since market introduction related to RISPERDAL overdose include prolonged QT interval and convulsions. Torsade de pointes has been reported in association with combined overdose of RISPERDAL and paroxetine.

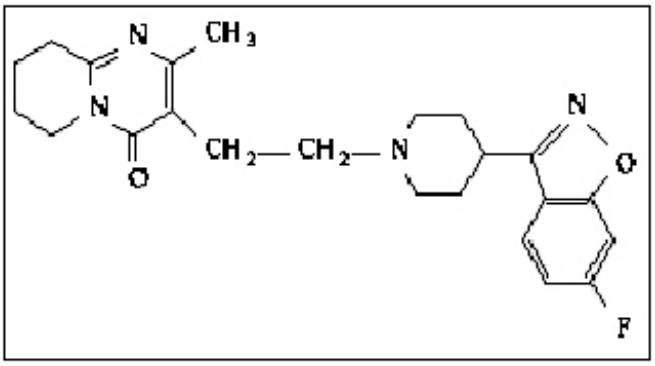
10.2 Management of Overdosage

For the most up to date information on the management of RISPERDAL overdosage, contact a certified poison control center (1-800-222-1222 or www.poison.org). Provide supportive care including close medical supervision and monitoring. Treatment should consist of general measures employed in the management of overdosage with any drug. Consider the possibility of multiple drug overdosage. Ensure an adequate airway, oxygenation, and ventilation. Monitor cardiac rhythm and vital signs. Use supportive and symptomatic measures. There is no specific antidote to RISPERDAL.

11 DESCRIPTION

RISPERDAL® contains risperidone, an atypical antipsychotic belonging to the chemical class of benzisoxazole derivatives. The chemical designation is 3-[2-[4-(6-fluoro-1,2-

benzisoxazol-3-yl)-1-piperidinyl]ethyl]-6,7,8,9-tetrahydro-2-methyl-4H-pyrido[1,2-a]pyrimidin-4-one. Its molecular formula is $C_{23}H_{27}FN_4O_2$ and its molecular weight is 410.49. The structural formula is:



Risperidone is a white to slightly beige powder. It is practically insoluble in water, freely soluble in methylene chloride, and soluble in methanol and 0.1 N HCl.

RISPERDAL® Tablets are for oral administration and available in 0.5 mg (red-brown), 1 mg (white), 2 mg (orange), 3 mg (yellow), and 4 mg (green) strengths. RISPERDAL tablets contain the following inactive ingredients: colloidal silicon dioxide, hypromellose, lactose, magnesium stearate, microcrystalline cellulose, propylene glycol, sodium lauryl sulfate, and starch (corn). The 0.5 mg, 2 mg, 3 mg, and 4 mg tablets also contain talc and titanium dioxide. The 0.5 mg tablets contain red iron oxide; the 2 mg tablets contain FD&C Yellow No. 6 Aluminum Lake; the 3 mg and 4 mg tablets contain D&C Yellow No. 10; the 4 mg tablets contain FD&C Blue No. 2 Aluminum Lake.

RISPERDAL is also available as a 1 mg/mL oral solution. RISPERDAL® Oral Solution contains the following inactive ingredients: tartaric acid, benzoic acid, sodium hydroxide, and purified water.

RISPERDAL® M-TAB® Orally Disintegrating Tablets are available in 0.5 mg (light coral), 1 mg (light coral), 2 mg (coral), 3 mg (coral), and 4 mg (coral) strengths. RISPERDAL M-TAB Orally Disintegrating Tablets contain the following inactive ingredients: Amberlite® resin, gelatin, mannitol, glycine, simethicone, carbomer, sodium hydroxide, aspartame, red ferric oxide, and peppermint oil. In addition, the 2 mg, 3 mg, and 4 mg RISPERDAL M-TAB Orally Disintegrating Tablets contain xanthan gum.

12 CLINICAL PHARMACOLOGY

12.1 Mechanism of Action

The mechanism of action of risperidone in schizophrenia is unclear. The drug's therapeutic activity in schizophrenia could be mediated through a combination of dopamine Type 2 (D_2) and serotonin Type 2 ($5HT_2$) receptor antagonism. The clinical effect from risperidone results from the combined concentrations of risperidone and its major metabolite, 9-hydroxyrisperidone (paliperidone) *[see Clinical Pharmacology (12.3)]*. Antagonism at receptors other than D_2 and $5HT_2$ may explain some of the other effects of risperidone *[see Clinical Pharmacology (12.1)]*.

12.2 Pharmacodynamics

Risperidone is a monoaminergic antagonist with high affinity (Ki of 0.12 to 7.3 nM) for the serotonin Type 2 ($5HT_2$), dopamine Type 2 (D_2), α_1 and α_2 adrenergic, and H_1 histaminergic receptors. Risperidone showed low to moderate affinity (Ki of 47 to 253 nM) for the serotonin $5HT_{1C}$, $5HT_{1D}$, and $5HT_{1A}$ receptors, weak affinity (Ki of 620 to 800 nM) for the dopamine D_1 and haloperidol-sensitive sigma site, and no affinity (when tested at concentrations $>10^{-5}$ M) for cholinergic muscarinic or β_1 and β_2 adrenergic receptors.

12.3 Pharmacokinetics

Absorption

Risperidone is well absorbed. The absolute oral bioavailability of risperidone is 70% (CV=25%). The relative oral bioavailability of risperidone from a tablet is 94% (CV=10%) when compared to a solution.

Pharmacokinetic studies showed that RISPERDAL M-TAB Orally Disintegrating Tablets and RISPERDAL Oral Solution are bioequivalent to RISPERDAL Tablets.

Plasma concentrations of risperidone, its major metabolite, 9-hydroxyrisperidone, and risperidone plus 9-hydroxyrisperidone are dose proportional over the dosing range of 1 to 16 mg daily (0.5 to 8 mg twice daily). Following oral administration of solution or tablet, mean peak plasma concentrations of risperidone occurred at about 1 hour. Peak concentrations of 9-hydroxyrisperidone occurred at about 3 hours in extensive metabolizers, and 17 hours in poor metabolizers. Steady-state concentrations of risperidone are reached in 1 day in extensive metabolizers and would be expected to reach steady-state in about 5 days in poor metabolizers. Steady-state concentrations of 9-hydroxyrisperidone are reached in 5–6 days (measured in extensive metabolizers).

Food Effect

Food does not affect either the rate or extent of absorption of risperidone. Thus, RISPERDAL can be given with or without meals.

Distribution

Risperidone is rapidly distributed. The volume of distribution is 1–2 L/kg. In plasma, risperidone is bound to albumin and α_1-acid glycoprotein. The plasma protein binding of risperidone is approximately 90%, and that of its major metabolite, 9-hydroxyrisperidone, is 77%. Neither risperidone nor 9-hydroxyrisperidone displaces each other from plasma binding sites. High therapeutic concentrations of sulfamethazine (100 mcg/mL), warfarin (10 mcg/mL), and carbamazepine (10mcg/mL) caused only a slight increase in the free fraction of risperidone at 10 ng/mL and 9-hydroxyrisperidone at 50 ng/mL, changes of unknown clinical significance.

Elimination

Metabolism

Risperidone is extensively metabolized in the liver. The main metabolic pathway is through hydroxylation of risperidone to 9-hydroxyrisperidone by the enzyme, CYP 2D6. A minor metabolic pathway is through *N*-dealkylation. The main metabolite, 9-hydroxyrisperidone, has similar pharmacological activity as risperidone. Consequently, the clinical effect of the drug results from the combined concentrations of risperidone

plus 9-hydroxyrisperidone.

CYP 2D6, also called debrisoquin hydroxylase, is the enzyme responsible for metabolism of many neuroleptics, antidepressants, antiarrhythmics, and other drugs. CYP 2D6 is subject to genetic polymorphism (about 6%–8% of Caucasians, and a very low percentage of Asians, have little or no activity and are "poor metabolizers") and to inhibition by a variety of substrates and some non-substrates, notably quinidine. Extensive CYP 2D6 metabolizers convert risperidone rapidly into 9-hydroxyrisperidone, whereas poor CYP 2D6 metabolizers convert it much more slowly. Although extensive metabolizers have lower risperidone and higher 9-hydroxyrisperidone concentrations than poor metabolizers, the pharmacokinetics of risperidone and 9-hydroxyrisperidone combined, after single and multiple doses, are similar in extensive and poor metabolizers.

Excretion

Risperidone and its metabolites are eliminated via the urine and, to a much lesser extent, via the feces. As illustrated by a mass balance study of a single 1 mg oral dose of ^{14}C-risperidone administered as solution to three healthy male volunteers, total recovery of radioactivity at 1 week was 84%, including 70% in the urine and 14% in the feces.

The apparent half-life of risperidone was 3 hours (CV=30%) in extensive metabolizers and 20 hours (CV=40%) in poor metabolizers. The apparent half-life of 9-hydroxyrisperidone was about 21 hours (CV=20%) in extensive metabolizers and 30 hours (CV=25%) in poor metabolizers. The pharmacokinetics of risperidone and 9-hydroxyrisperidone combined, after single and multiple doses, were similar in extensive and poor metabolizers, with an overall mean elimination half-life of about 20 hours.

Drug Interaction Studies

Risperidone could be subject to two kinds of drug-drug interactions. First, inhibitors of CYP 2D6 interfere with conversion of risperidone to 9-hydroxyrisperidone *[see Drug Interactions (7)]*. This occurs with quinidine, giving essentially all recipients a risperidone pharmacokinetic profile typical of poor metabolizers. The therapeutic benefits and adverse effects of risperidone in patients receiving quinidine have not been evaluated, but observations in a modest number (n≅70) of poor metabolizers given RISPERDAL do not suggest important differences between poor and extensive metabolizers. Second, co-administration of known enzyme inducers (e.g., carbamazepine, phenytoin, rifampin, and phenobarbital) with RISPERDAL may cause a decrease in the combined plasma concentrations of risperidone and 9-hydroxyrisperidone *[see Drug Interactions (7)]*. It would also be possible for risperidone to interfere with metabolism of other drugs metabolized by CYP 2D6. Relatively weak binding of risperidone to the enzyme suggests this is unlikely *[see Drug Interactions (7)]*.

In vitro studies indicate that risperidone is a relatively weak inhibitor of CYP 2D6. Therefore, RISPERDAL is not expected to substantially inhibit the clearance of drugs that are metabolized by this enzymatic pathway. In drug interaction studies, RISPERDAL did not significantly affect the pharmacokinetics of donepezil and galantamine, which are metabolized by CYP 2D6.

In vitro studies demonstrated that drugs metabolized by other CYP isozymes, including 1A1, 1A2, 2C9, 2C19, and 3A4, are only weak inhibitors of risperidone metabolism.

Specific Populations

Renal and Hepatic Impairment

[See Use in Specific Populations (8.6 and 8.7)].

Elderly

In healthy elderly subjects, renal clearance of both risperidone and 9-hydroxyrisperidone was decreased, and elimination half-lives were prolonged compared to young healthy subjects. Dosing should be modified accordingly in the elderly patients *[see Use in Specific Populations (8.5)]*.

Pediatric

The pharmacokinetics of risperidone and 9-hydroxyrisperidone in children were similar to those in adults after correcting for the difference in body weight.

Race and Gender Effects

No specific pharmacokinetic study was conducted to investigate race and gender effects, but a population pharmacokinetic analysis did not identify important differences in the disposition of risperidone due to gender (whether corrected for body weight or not) or race.

13 NONCLINICAL TOXICOLOGY

13.1 Carcinogenesis, Mutagenesis, Impairment of Fertility

Carcinogenesis

Risperidone was administered in the diet at doses of 0.63, 2.5, and 10 mg/kg for 18 months to mice and for 25 months to rats. These doses are equivalent to approximately 0.2, 0.75, and 3 times (mice) and 0.4, 1.5, and 6 times (rats) the MRHD of 16 mg/day, based on mg/m^2 body surface area. A maximum tolerated dose was not achieved in male mice. There were statistically significant increases in pituitary gland adenomas, endocrine pancreas adenomas, and mammary gland adenocarcinomas. The table below summarizes the multiples of the human dose on a mg/m^2 (mg/kg) basis at which these tumors occurred.

			Multiples of Maximum Human Dose in mg/m^2 (mg/kg)	
Tumor Type	**Species**	**Sex**	**Lowest Effect Level**	**Highest No-Effect Level**
Pituitary adenomas	mouse	Female	0.75 (9.4)	0.2 (2.4)
Endocrine pancreas adenomas	rat	Male	1.5 (9.4)	0.4 (2.4)
Mammary gland adenocarcinomas	mouse	Female	0.2 (2.4)	none
	rat	Female	0.4 (2.4)	none
	rat	Male	6.0 (37.5)	1.5 (9.4)
Mammary gland neoplasm,	rat	Male	1.5 (9.4)	0.4 (2.4)

Total	Rat	Male	1.5 (9.4)	0.4 (2.4)

Antipsychotic drugs have been shown to chronically elevate prolactin levels in rodents. Serum prolactin levels were not measured during the risperidone carcinogenicity studies; however, measurements during subchronic toxicity studies showed that risperidone elevated serum prolactin levels 5–6 fold in mice and rats at the same doses used in the carcinogenicity studies. An increase in mammary, pituitary, and endocrine pancreas neoplasms has been found in rodents after chronic administration of other antipsychotic drugs and is considered to be prolactin-mediated. The relevance for human risk of the findings of prolactin-mediated endocrine tumors in rodents is unclear *[see Warnings and Precautions (5.6)]*.

Mutagenesis

No evidence of mutagenic or clastogenic potential for risperidone was found in the *in vitro* tests of Ames gene mutation, the mouse lymphoma assay, rat hepatocyte DNA-repair assay, the chromosomal aberration test in human lymphocytes, Chinese hamster ovary cells, or in the *in vivo* oral micronucleus test in mice and the sex-linked recessive lethal test in *Drosophila*.

Impairment of Fertility

Oral risperidone (0.16 to 5 mg/kg) impaired mating, but not fertility, in rat reproductive studies at doses 0.1 to 3 times the MRHD of 16 mg/day based on mg/m^2 body surface area. The effect appeared to be in females, since impaired mating behavior was not noted in the male fertility study. In a subchronic study in Beagle dogs in which risperidone was administered orally at doses of 0.31 to 5 mg/kg, sperm motility and concentration were decreased at doses 0.6 to 10 times the MRHD based on mg/m^2 body surface area. Dose-related decreases were also noted in serum testosterone at the same doses. Serum testosterone and sperm parameters partially recovered, but remained decreased after treatment was discontinued. A no-effect dose could not be determined in either rat or dog.

14 CLINICAL STUDIES

14.1 Schizophrenia

Adults

Short-Term Efficacy

The efficacy of RISPERDAL in the treatment of schizophrenia was established in four short-term (4- to 8-week) controlled trials of psychotic inpatients who met DSM-III-R criteria for schizophrenia.

Several instruments were used for assessing psychiatric signs and symptoms in these studies, among them the Brief Psychiatric Rating Scale (BPRS), a multi-item inventory of general psychopathology traditionally used to evaluate the effects of drug treatment in schizophrenia. The BPRS psychosis cluster (conceptual disorganization, hallucinatory behavior, suspiciousness, and unusual thought content) is considered a particularly useful subset for assessing actively psychotic schizophrenic patients. A second traditional assessment, the Clinical Global Impression (CGI), reflects the impression of a

skilled observer, fully familiar with the manifestations of schizophrenia, about the overall clinical state of the patient. In addition, the Positive and Negative Syndrome Scale (PANSS) and the Scale for Assessing Negative Symptoms (SANS) were employed.

The results of the trials follow:

(1) In a 6-week, placebo-controlled trial (n=160) involving titration of RISPERDAL in doses up to 10 mg/day (twice-daily schedule), RISPERDAL was generally superior to placebo on the BPRS total score, on the BPRS psychosis cluster, and marginally superior to placebo on the SANS.
(2) In an 8-week, placebo-controlled trial (n=513) involving 4 fixed doses of RISPERDAL (2 mg/day, 6 mg/day, 10 mg/day, and 16 mg/day, on a twice-daily schedule), all 4 RISPERDAL groups were generally superior to placebo on the BPRS total score, BPRS psychosis cluster, and CGI severity score; the 3 highest RISPERDAL dose groups were generally superior to placebo on the PANSS negative subscale. The most consistently positive responses on all measures were seen for the 6 mg dose group, and there was no suggestion of increased benefit from larger doses.
(3) In an 8-week, dose comparison trial (n=1356) involving 5 fixed doses of RISPERDAL (1 mg/day, 4 mg/day, 8 mg/day, 12 mg/day, and 16 mg/day, on a twice-daily schedule), the four highest RISPERDAL dose groups were generally superior to the 1 mg RISPERDAL dose group on BPRS total score, BPRS psychosis cluster, and CGI severity score. None of the dose groups were superior to the 1 mg group on the PANSS negative subscale. The most consistently positive responses were seen for the 4 mg dose group.
(4) In a 4-week, placebo-controlled dose comparison trial (n=246) involving 2 fixed doses of RISPERDAL (4 and 8 mg/day on a once-daily schedule), both RISPERDAL dose groups were generally superior to placebo on several PANSS measures, including a response measure (>20% reduction in PANSS total score), PANSS total score, and the BPRS psychosis cluster (derived from PANSS). The results were generally stronger for the 8 mg than for the 4 mg dose group.

Long-Term Efficacy

In a longer-term trial, 365 adult outpatients predominantly meeting DSM-IV criteria for schizophrenia and who had been clinically stable for at least 4 weeks on an antipsychotic medication were randomized to RISPERDAL (2–8 mg/day) or to an active comparator, for 1 to 2 years of observation for relapse. Patients receiving RISPERDAL experienced a significantly longer time to relapse over this time period compared to those receiving the active comparator.

Pediatrics

The efficacy of RISPERDAL in the treatment of schizophrenia in adolescents aged 13–17 years was demonstrated in two short-term (6 and 8 weeks), double-blind controlled trials. All patients met DSM-IV diagnostic criteria for schizophrenia and were experiencing an acute episode at time of enrollment. In the first trial (study #1), patients were randomized into one of three treatment groups: RISPERDAL 1–3 mg/day (n=55, mean modal dose = 2.6 mg), RISPERDAL 4–6 mg/day (n=51, mean modal dose = 5.3 mg), or placebo (n=54). In the second trial (study #2), patients were randomized to either RISPERDAL 0.15–0.6 mg/day (n=132, mean modal dose = 0.5 mg) or RISPERDAL 1.5–6 mg/day (n=125, mean modal dose = 4 mg). In all cases, study medication was initiated at 0.5 mg/day (with the exception of the 0.15–0.6 mg/day group in study #2,

where the initial dose was 0.05 mg/day) and titrated to the target dosage range by approximately Day 7. Subsequently, dosage was increased to the maximum tolerated dose within the target dose range by Day 14. The primary efficacy variable in all studies was the mean change from baseline in total PANSS score.

Results of the studies demonstrated efficacy of RISPERDAL in all dose groups from 1–6 mg/day compared to placebo, as measured by significant reduction of total PANSS score. The efficacy on the primary parameter in the 1–3 mg/day group was comparable to the 4–6 mg/day group in study #1, and similar to the efficacy demonstrated in the 1.5–6 mg/day group in study #2. In study #2, the efficacy in the 1.5–6 mg/day group was statistically significantly greater than that in the 0.15–0.6 mg/day group. Doses higher than 3 mg/day did not reveal any trend towards greater efficacy.

14.2 Bipolar Mania - Monotherapy

Adults

The efficacy of RISPERDAL in the treatment of acute manic or mixed episodes was established in two short-term (3-week) placebo-controlled trials in patients who met the DSM-IV criteria for Bipolar I Disorder with manic or mixed episodes. These trials included patients with or without psychotic features.

The primary rating instrument used for assessing manic symptoms in these trials was the Young Mania Rating Scale (YMRS), an 11-item clinician-rated scale traditionally used to assess the degree of manic symptomatology (irritability, disruptive/aggressive behavior, sleep, elevated mood, speech, increased activity, sexual interest, language/thought disorder, thought content, appearance, and insight) in a range from 0 (no manic features) to 60 (maximum score). The primary outcome in these trials was change from baseline in the YMRS total score. The results of the trials follow:

(1) In one 3-week placebo-controlled trial (n=246), limited to patients with manic episodes, which involved a dose range of RISPERDAL 1–6 mg/day, once daily, starting at 3 mg/day (mean modal dose was 4.1 mg/day), RISPERDAL was superior to placebo in the reduction of YMRS total score.
(2) In another 3-week placebo-controlled trial (n=286), which involved a dose range of 1–6 mg/day, once daily, starting at 3 mg/day (mean modal dose was 5.6 mg/day), RISPERDAL was superior to placebo in the reduction of YMRS total score.

Pediatrics

The efficacy of RISPERDAL in the treatment of mania in children or adolescents with Bipolar I disorder was demonstrated in a 3-week, randomized, double-blind, placebo-controlled, multicenter trial including patients ranging in ages from 10 to 17 years who were experiencing a manic or mixed episode of bipolar I disorder. Patients were randomized into one of three treatment groups: RISPERDAL 0.5–2.5 mg/day (n=50, mean modal dose = 1.9 mg), RISPERDAL 3–6 mg/day (n=61, mean modal dose = 4.7 mg), or placebo (n=58). In all cases, study medication was initiated at 0.5 mg/day and titrated to the target dosage range by Day 7, with further increases in dosage to the maximum tolerated dose within the targeted dose range by Day 10. The primary rating instrument used for assessing efficacy in this study was the mean change from baseline in the total YMRS score.

Results of this study demonstrated efficacy of RISPERDAL in both dose groups

compared with placebo, as measured by significant reduction of total YMRS score. The efficacy on the primary parameter in the 3–6 mg/day dose group was comparable to the 0.5–2.5 mg/day dose group. Doses higher than 2.5 mg/day did not reveal any trend towards greater efficacy.

14.3 Bipolar Mania – Adjunctive Therapy with Lithium or Valproate

The efficacy of RISPERDAL with concomitant lithium or valproate in the treatment of acute manic or mixed episodes was established in one controlled trial in adult patients who met the DSM-IV criteria for Bipolar I Disorder. This trial included patients with or without psychotic features and with or without a rapid-cycling course.

(1) In this 3-week placebo-controlled combination trial, 148 in- or outpatients on lithium or valproate therapy with inadequately controlled manic or mixed symptoms were randomized to receive RISPERDAL, placebo, or an active comparator, in combination with their original therapy. RISPERDAL, in a dose range of 1–6 mg/day, once daily, starting at 2 mg/day (mean modal dose of 3.8 mg/day), combined with lithium or valproate (in a therapeutic range of 0.6 mEq/L to 1.4 mEq/L or 50 mcg/mL to 120 mcg/mL, respectively) was superior to lithium or valproate alone in the reduction of YMRS total score.

(2) In a second 3-week placebo-controlled combination trial, 142 in- or outpatients on lithium, valproate, or carbamazepine therapy with inadequately controlled manic or mixed symptoms were randomized to receive RISPERDAL or placebo, in combination with their original therapy. RISPERDAL, in a dose range of 1–6 mg/day, once daily, starting at 2 mg/day (mean modal dose of 3.7 mg/day), combined with lithium, valproate, or carbamazepine (in therapeutic ranges of 0.6 mEq/L to 1.4 mEq/L for lithium, 50 mcg/mL to 125 mcg/mL for valproate, or 4–12 mcg/mL for carbamazepine, respectively) was not superior to lithium, valproate, or carbamazepine alone in the reduction of YMRS total score. A possible explanation for the failure of this trial was induction of risperidone and 9-hydroxyrisperidone clearance by carbamazepine, leading to subtherapeutic levels of risperidone and 9-hydroxyrisperidone.

14.4 Irritability Associated with Autistic Disorder

Short-Term Efficacy

The efficacy of RISPERDAL in the treatment of irritability associated with autistic disorder was established in two 8-week, placebo-controlled trials in children and adolescents (aged 5 to 16 years) who met the DSM-IV criteria for autistic disorder. Over 90% of these subjects were under 12 years of age and most weighed over 20 kg (16–104.3 kg).

Efficacy was evaluated using two assessment scales: the Aberrant Behavior Checklist (ABC) and the Clinical Global Impression - Change (CGI-C) scale. The primary outcome measure in both trials was the change from baseline to endpoint in the Irritability subscale of the ABC (ABC-I). The ABC-I subscale measured the emotional and behavioral symptoms of autism, including aggression towards others, deliberate self-injuriousness, temper tantrums, and quickly changing moods. The CGI-C rating at endpoint was a co-primary outcome measure in one of the studies.

The results of these trials are as follows:

(1) In one of the 8-week, placebo-controlled trials, children and adolescents with

autistic disorder (n=101), aged 5 to 16 years, received twice daily doses of placebo or RISPERDAL 0.5–3.5 mg/day on a weight-adjusted basis. RISPERDAL, starting at 0.25 mg/day or 0.5 mg/day depending on baseline weight (< 20 kg and ≥ 20 kg, respectively) and titrated to clinical response (mean modal dose of 1.9 mg/day, equivalent to 0.06 mg/kg/day), significantly improved scores on the ABC-I subscale and on the CGI-C scale compared with placebo.

(2) In the other 8-week, placebo-controlled trial in children with autistic disorder (n=55), aged 5 to 12 years, RISPERDAL 0.02 to 0.06 mg/kg/day given once or twice daily, starting at 0.01 mg/kg/day and titrated to clinical response (mean modal dose of 0.05 mg/kg/day, equivalent to 1.4 mg/day), significantly improved scores on the ABC-I subscale compared with placebo.

A third trial was a 6-week, multicenter, randomized, double-blind, placebo-controlled, fixed-dose study to evaluate the efficacy and safety of a lower than recommended dose of risperidone in subjects (N=96) 5 to 17 years of age with autistic disorder (defined by DSM-IV criteria) and associated irritability and related behavioral symptoms. Approximately 77% of patients were younger than 12 years of age (mean age = 9), and 88% were male. Most patients (73%) weighed less than 45 kg (mean weight = 40 kg). Approximately 90% of patients were antipsychotic-naïve before entering the study.

There were two weight-based, fixed doses of risperidone (high-dose and low-dose). The high dose was 1.25 mg per day for patients weighing 20 to < 45 kg, and it was 1.75 mg per day for patients weighing ≥ 45 kg. The low dose was 0.125 mg per day for patients weighing 20 to < 45 kg, and it was 0.175 mg per day for patients weighing ≥ 45 kg. The dose was administered once daily in the morning, or in the evening if sedation occurred.

The primary efficacy endpoint was the mean change in the Aberrant Behavior Checklist – Irritability subscale (ABC-I) score from baseline to the end of Week 6. The study demonstrated the efficacy of high-dose risperidone, as measured by the mean change in ABC-I score. It did not demonstrate efficacy for low-dose risperidone. The mean baseline ABC-I scores were 29 in the placebo group (n=35), 27 in the risperidone low-dose group (n=30), and 28 in the risperidone high-dose group (n=31). The mean changes in ABC-I scores were -3.5, -7.4, and -12.4 in the placebo, low-dose, and high-dose group respectively. The results in the high-dose group were statistically significant ($p< 0.001$) but not in the low-dose group ($p=0.164$).

Long-Term Efficacy

Following completion of the first 8-week double-blind study, 63 patients entered an open-label study extension where they were treated with RISPERDAL for 4 or 6 months (depending on whether they received RISPERDAL or placebo in the double-blind study). During this open-label treatment period, patients were maintained on a mean modal dose of RISPERDAL of 1.8–2.1 mg/day (equivalent to 0.05 – 0.07 mg/kg/day).

Patients who maintained their positive response to RISPERDAL (response was defined as ≥ 25% improvement on the ABC-I subscale and a CGI-C rating of 'much improved' or 'very much improved') during the 4–6 month open-label treatment phase for about 140 days, on average, were randomized to receive RISPERDAL or placebo during an 8-week, double-blind withdrawal study (n=39 of the 63 patients). A pre-planned interim analysis of data from patients who completed the withdrawal study (n=32), undertaken by an independent Data Safety Monitoring Board, demonstrated a significantly lower relapse rate in the RISPERDAL group compared with the placebo group. Based on the interim analysis results, the study was terminated due to demonstration of a statistically

significant effect on relapse prevention. Relapse was defined as ≥ 25% worsening on the most recent assessment of the ABC-I subscale (in relation to baseline of the randomized withdrawal phase).

16 HOW SUPPLIED/STORAGE AND HANDLING

16.1 How Supplied

RISPERDAL® (risperidone) Tablets

RISPERDAL® (risperidone) Tablets are imprinted "JANSSEN" on one side and either "Ris 0.5", "R1", "R2", "R3", or "R4" according to their respective strengths.

0.5 mg red-brown, capsule-shaped tablets: bottles of 60 NDC 50458-302-06, bottles of 500 NDC 50458-302-50, and hospital unit dose blister packs of 100 NDC 50458-302-01.

1 mg white, capsule-shaped tablets: bottles of 60 NDC 50458-300-06, bottles of 500 NDC 50458-300-50, and hospital unit dose blister packs of 100 NDC 50458-300-01.

2 mg orange, capsule-shaped tablets: bottles of 60 NDC 50458-320-06, bottles of 500 NDC 50458-320-50, and hospital unit dose blister packs of 100 NDC 50458-320-01.

3 mg yellow, capsule-shaped tablets: bottles of 60 NDC 50458-330-06, bottles of 500 NDC 50458-330-50, and hospital unit dose blister packs of 100 NDC 50458-330-01.

4 mg green, capsule-shaped tablets: bottles of 60 NDC 50458-350-06 and hospital unit dose blister packs of 100 NDC 50458-350-01.

RISPERDAL® (risperidone) Oral Solution

RISPERDAL® (risperidone) 1 mg/mL Oral Solution (NDC 50458-305-03) is supplied in 30 mL bottles with a calibrated (in milliliters) oral dosing syringe. The minimum calibrated volume is 0.25 mL, while the maximum calibrated volume is 3 mL.

RISPERDAL® M-TAB® (risperidone) Orally Disintegrating Tablets

RISPERDAL® M-TAB® (risperidone) Orally Disintegrating Tablets are etched on one side with "R0.5", "R1", "R2", "R3", or "R4" according to their respective strengths. RISPERDAL M-TAB Orally Disintegrating Tablets 0.5 mg, 1 mg, and 2 mg are packaged in blister packs of 4 (2 × 2) tablets. Orally Disintegrating Tablets 3 mg and 4 mg are packaged in a child-resistant pouch containing a blister with 1 tablet.

0.5 mg light coral, round, biconvex tablets: 7 blister packages (4 tablets each) per box, NDC 50458-395-28, and long-term care blister packaging of 30 tablets NDC 50458-395-30.

1 mg light coral, square, biconvex tablets: 7 blister packages (4 tablets each) per box, NDC 50458-315-28, and long-term care blister packaging of 30 tablets NDC 50458-315-30.

2 mg coral, square, biconvex tablets: 7 blister packages (4 tablets each) per box, NDC 50458-325-28.

3 mg coral, round, biconvex tablets: 28 blisters per box, NDC 50458-335-28.

4 mg coral, round, biconvex tablets: 28 blisters per box, NDC 50458-355-28.

16.2 Storage and Handling

RISPERDAL Tablets should be stored at controlled room temperature 15 °C to 25 °C (59 °F to 77 °F). Protect from light and moisture.

RISPERDAL 1 mg/mL Oral Solution should be stored at controlled room temperature 15 °C to 25 °C (59 °F to 77 °F). Protect from light and freezing.

RISPERDAL M-TAB Orally Disintegrating Tablets should be stored at controlled room temperature 15 °C to 25 °C (59 °F to 77 °F).

Keep out of reach of children.

17 PATIENT COUNSELING INFORMATION

Advise patients using RISPERDAL oral solution to read the FDA-approved patient labeling (Instructions for Use) for RISPERDAL oral solution.

Physicians are advised to discuss the following issues with patients for whom they prescribe RISPERDAL.

Neuroleptic Malignant Syndrome (NMS)

Counsel patients about a potentially fatal adverse reaction, Neuroleptic Malignant Syndrome (NMS), that has been reported in association with administration of antipsychotic drugs. Advise patients, family members, or caregivers to contact the healthcare provider or report to the emergency room if they experience signs and symptoms of NMS, including hyperpyrexia, muscle rigidity, altered mental status including delirium, and evidence of autonomic instability (irregular pulse or blood pressure, tachycardia, diaphoresis, and cardiac dysrhythmia) *[see Warnings and Precautions (5.3)]*.

Tardive Dyskinesia

Counsel patients on the signs and symptoms of tardive dyskinesia and to contact their healthcare provider if these abnormal movements occur *[see Warnings and Precautions (5.4)]*.

Metabolic Changes

Educate patients about the risk of metabolic changes, how to recognize symptoms of hyperglycemia and diabetes mellitus, and the need for specific monitoring, including blood glucose, lipids, and weight *[see Warnings and Precautions (5.5)]*.

Orthostatic Hypotension

Educate patients about the risk of orthostatic hypotension and syncope, particularly at the time of initiating treatment, re-initiating treatment, or increasing the dose *[see Warnings and Precautions (5.7)]*.

Leukopenia/Neutropenia

Advise patients with a pre-existing low WBC or a history of drug induced leukopenia/neutropenia they should have their CBC monitored while taking RISPERDAL *[see Warnings and Precautions (5.9)]*.

Hyperprolactinemia

Counsel patients on signs and symptoms of hyperprolactinemia that may be associated with chronic use of RISPERDAL. Advise them to seek medical attention if they experience any of the following: amenorrhea or galactorrhea in females, erectile dysfunction or gynecomastia in males. *[see Warnings and Precautions (5.6)]*.

Interference with Cognitive and Motor Performance

Caution patients about performing activities requiring mental alertness, such as operating hazardous machinery, or operating a motor vehicle until they are reasonably certain that RISPERDAL therapy does not affect them adversely *[see Warnings and Precautions (5.10)]*.

Priapism

Advise patients of the possibility of painful or prolonged penile erections (priapism). Instruct the patient to seek immediate medical attention in the event of priapism *[see Warnings and Precautions (5.13)]*.

Heat Exposure and Dehydration

Counsel patients regarding appropriate care in avoiding overheating and dehydration *[see Warnings and Precautions (5.14)]*.

Phenylketonurics

Inform patients with Phenylketonuria and caregivers that RISPERDAL M-TAB Orally Disintegrating Tablets contain phenylalanine. Phenylalanine is a component of aspartame. Each 4 mg RISPERDAL M-TAB Orally Disintegrating Tablet contains 0.84 mg phenylalanine; each 3 mg RISPERDAL M-TAB Orally Disintegrating Tablet contains 0.63 mg phenylalanine; each 2 mg RISPERDAL M-TAB Orally Disintegrating Tablet contains 0.42 mg phenylalanine; each 1 mg RISPERDAL M-TAB Orally Disintegrating Tablet contains 0.28 mg phenylalanine; and each 0.5 mg RISPERDAL M-TAB Orally Disintegrating Tablet contains 0.14 mg phenylalanine *[see Warnings and Precautions (5.15)]*.

Concomitant Medication

Advise patients to inform their healthcare providers if they are taking, or plan to take any prescription or over-the-counter drugs, as there is a potential for interactions *[see Drug Interactions (7)]*.

Alcohol

Advise patients to avoid alcohol while taking RISPERDAL *[see Drug Interactions (7.2)]*.

Pregnancy

Advise patients to notify their healthcare provider if they become pregnant or intend to become pregnant during treatment with RISPERDAL. Advise patients that RISPERDAL may cause extrapyramidal and/or withdrawal symptoms in a neonate. Advise patients that there is a pregnancy registry that monitors pregnancy outcomes in women exposed to RISPERDAL during pregnancy *[see Use in Specific Populations (8.1)]*.

Lactation

Advise breastfeeding women using RISPERDAL to monitor infants for somnolence, failure to thrive, jitteriness, and extrapyramidal symptoms (tremors and abnormal muscle movements) and to seek medical care if they notice these signs *[see Use in*

Specific Populations (8.2)].

Infertility

Advise females of reproductive potential that RISPERDAL may impair fertility due to an increase in serum prolactin levels. The effects on fertility are reversible *[see Use in Specific Populations (8.3)].*

RISPERDAL Tablets, RISPERDAL M-TAB Orally Disintegrating Tablets, and RISPERDAL Oral Solution are manufactured for:
Janssen Pharmaceuticals, Inc.
Titusville, NJ 08560, USA

For patent information: www.janssenpatents.com

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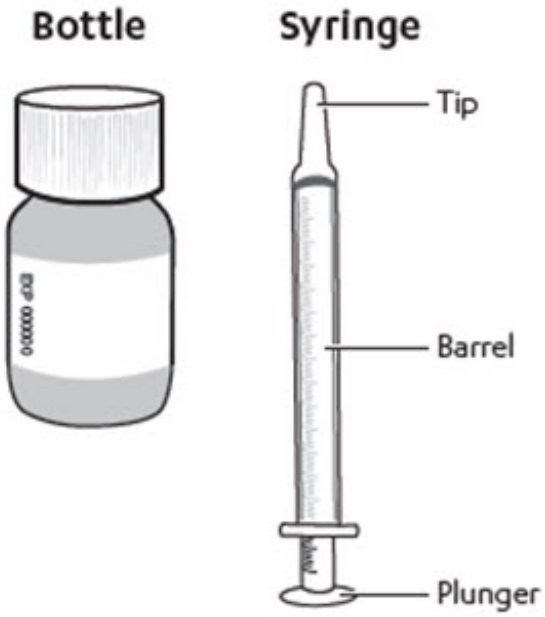
INSTRUCTIONS FOR USE

RISPERDAL ® (RISS-per-dal)
(risperidone)
Oral Solution

Read these Instructions for Use before you start using RISPERDAL Oral Solution and each time you get a refill. There may be new information. This information does not take the place of talking to your healthcare provider about your medical condition or treatment.

Important information you need to know before taking RISPERDAL Oral Solution:

- Take RISPERDAL Oral Solution exactly as your healthcare provider tells you to take it.
- Each 1 mL contains 1 mg of RISPERDAL Oral Solution.
- Ask your healthcare provider or pharmacist to show you how to measure your prescribed dose using the oral dosing syringe.
- Always use the oral dosing syringe that comes with RISPERDAL Oral Solution. Contact your healthcare provider or pharmacist if you lose or damage the oral dosing syringe, or if your carton does not come with one.
- RISPERDAL Oral Solution can be taken directly from the oral dosing syringe or mixed with water, coffee, orange juice, or low-fat milk. **Do not** mix RISPERDAL Oral Solution with cola or tea.



Each RISPERDAL Oral Solution carton contains:

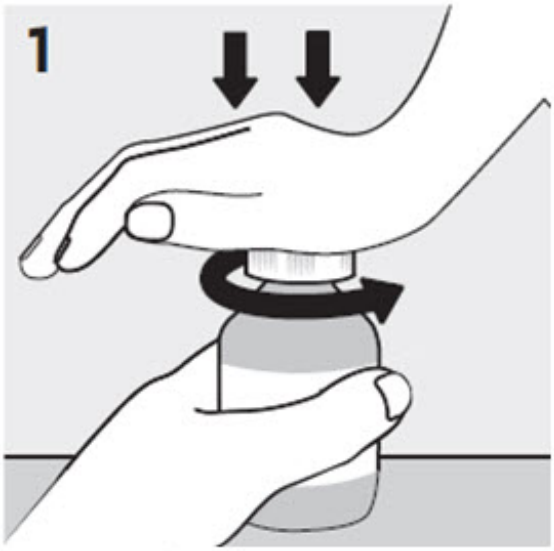
- 1 bottle of RISPERDAL Oral Solution
- 1 oral dosing syringe

Gather and check supplies:

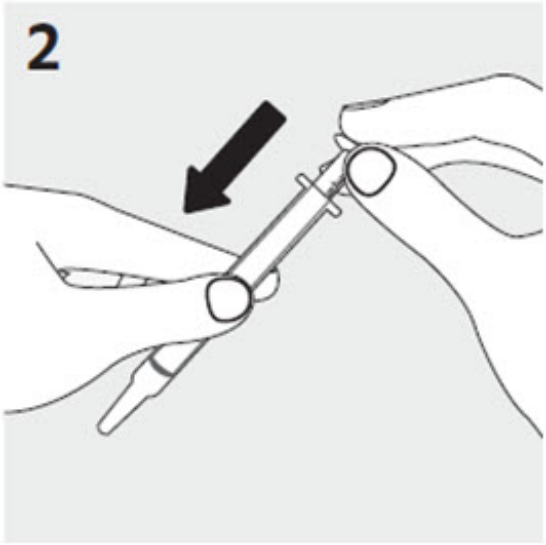
- Gather the RISPERDAL Oral Solution bottle and oral dosing syringe.
- Check the expiration date on the bottle. Do not use the bottle of RISPERDAL Oral Solution if the expiration date has passed.
- Check your dose in mLs as prescribed by your healthcare provider. Find this mL marking on the plunger of the oral dosing syringe. If your dose is more than 3 mL, you will need to divide your dose. Follow the instructions given to you by your healthcare provider or pharmacist on how to divide your dose.

Preparing a dose of RISPERDAL Oral Solution:

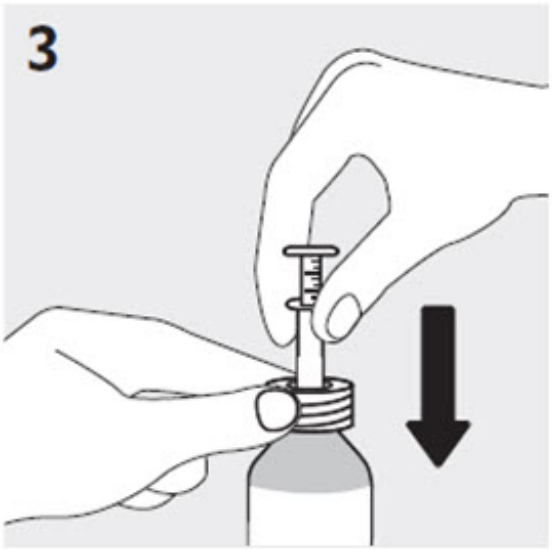
Step 1. Place the RISPERDAL Oral Solution bottle on a flat surface. Push down on the cap while turning it to the left (counterclockwise) to open the bottle.



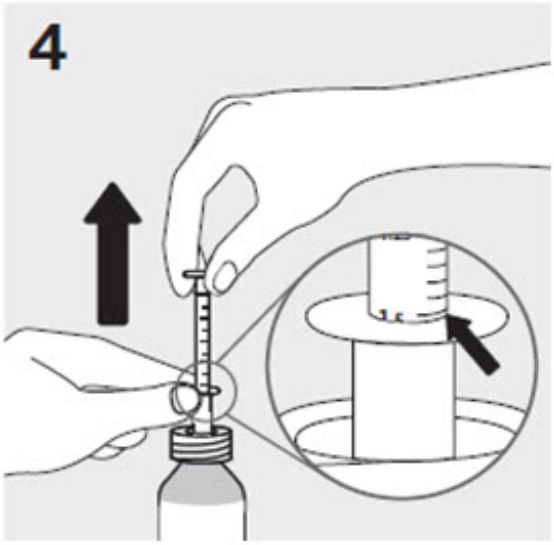
Step 2. Push the plunger of the oral dosing syringe all the way down.



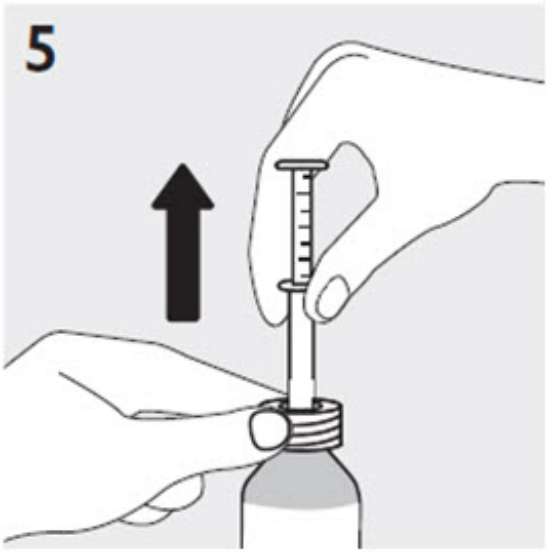
Step 3. With the bottle in an upright position, fully insert the oral dosing syringe into the opening of the bottle.



Step 4. Withdraw the prescribed dose of RISPERDAL Oral Solution from the bottle. Hold down the barrel of the oral dosing syringe with one hand. With your other hand, slowly pull the plunger up until you reach the mL markings on the plunger for the prescribed dose.

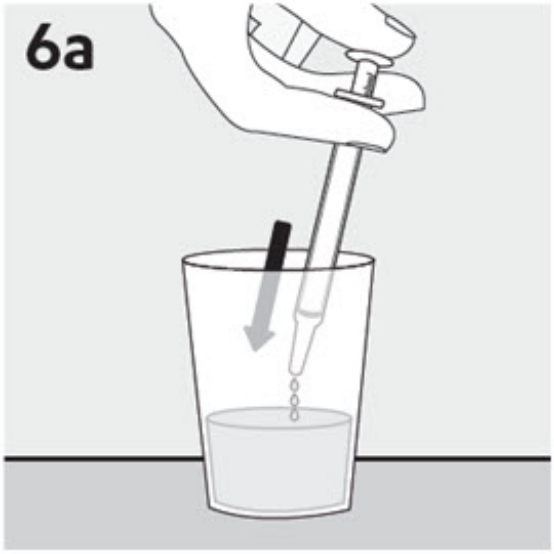


Step 5. Remove the oral dosing syringe from the bottle by holding the outer barrel and pulling straight up. Be careful not to push down on the plunger during this step.
Check the oral dosing syringe for air bubbles. If you see air bubbles, slowly push the plunger all the way down to return the oral solution into the bottle. Then repeat **Step 4** to withdraw the prescribed dose.



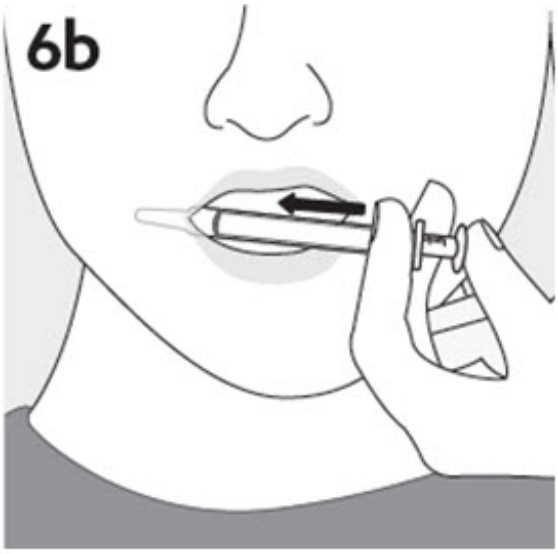
Step 6. RISPERDAL Oral Solution can be mixed with a drink or taken directly from the oral dosing syringe.

- Mix the dose of RISPERDAL Oral Solution with water, coffee, orange juice, or low-fat milk. Stir well and drink all of the mixture right away to ensure the full dose is taken. **See Figure 6a. Do not mix** RISPERDAL Oral Solution with cola or tea.

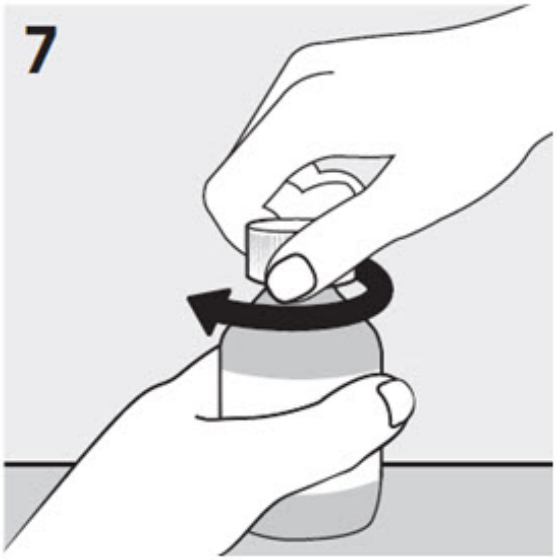


Or

- To take the RISPERDAL Oral Solution dose directly from the oral dosing syringe, place the tip of the oral dosing syringe into the mouth and toward the cheek. Slowly push the plunger all the way down to gently release all of the medicine in the oral dosing syringe. Do not squirt or forcefully push the medicine into the back of the throat. **See Figure 6b**.



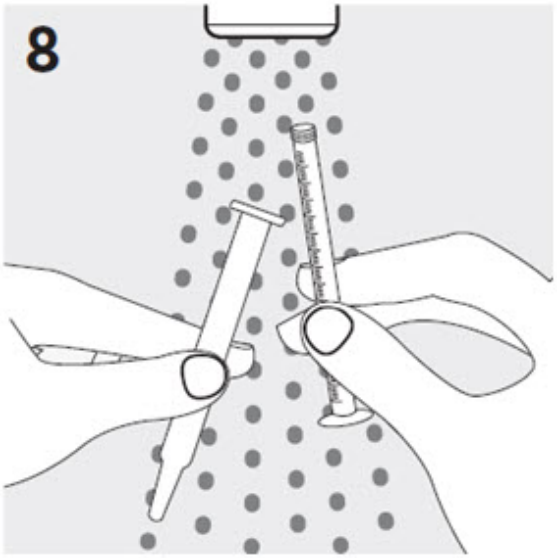
Step 7. Place the cap back on the RISPERDAL Oral Solution bottle and turn the cap to the right (clockwise) to close the bottle.



Step 8. Rinse the oral dosing syringe with water after each use.

- Remove the plunger from the oral dosing syringe barrel.
- Rinse the oral dosing syringe barrel and plunger with water and let them air dry.
- When the oral dosing syringe barrel and plunger are dry, put the plunger back into the oral dosing syringe barrel for the next use.

Do not throw away the oral dosing syringe.



Storing RISPERDAL Oral Solution:

- Store RISPERDAL Oral Solution at room temperature between 59 °F to 77 °F (15 °C to 25 °C).
- Do not freeze RISPERDAL Oral Solution. Protect from light.
- **Keep RISPERDAL Oral Solution and all medicines out of the reach of children.**

Manufactured for:
Janssen Pharmaceuticals, Inc.
Titusville, NJ, 08560, USA

For patent information: www.jansenpatents.com

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This Instructions for Use has been approved by the U.S. Food and Drug Administration.
Issued: 2/2025

INSTRUCCIONES DE USO

RISPERDAL ®(RISS-per-dal)

(risperidona)
Solución oral

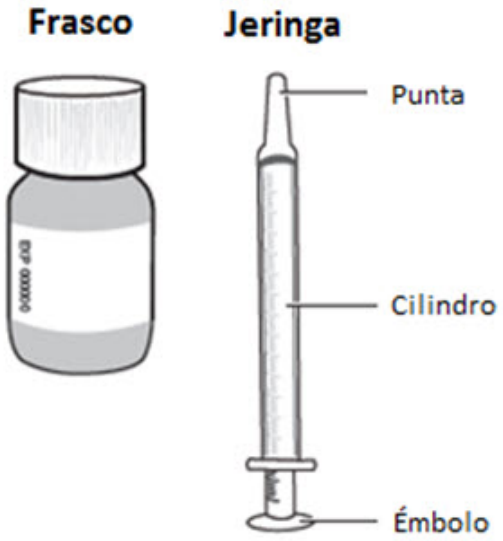
Lea estas Instrucciones de uso antes de comenzar a usar RISPERDAL Solución oral y cada vez que renueve su receta. Es posible que este material contenga información nueva. Esta información no reemplaza la consulta con su proveedor de atención médica acerca de su enfermedad o tratamiento.

Información importante que debe saber antes de tomar RISPERDAL Solución oral:

- Tome RISPERDAL Solución oral exactamente como se lo indica su proveedor de atención médica.
- Cada 1 ml contiene 1 mg de RISPERDAL Solución oral.
- Pídale a su proveedor de atención médica o farmacéutico que le muestre cómo medir la dosis recetada con la jeringa de dosificación oral.
- Utilice siempre la jeringa de dosificación oral que viene con RISPERDAL Solución oral. Comuníquese con su proveedor de atención médica o farmacéutico si pierde o daña la jeringa de dosificación oral, o si su caja no viene con una.
- RISPERDAL Solución oral se puede tomar directamente de la jeringa de dosificación oral o se puede mezclar con agua, café, jugo de naranja o leche con bajo contenido de grasa. **No** mezcle RISPERDAL Solución oral con refrescos de cola o té.

Cada caja de RISPERDAL Solución oral contiene lo siguiente:

- 1 frasco de RISPERDAL Solución oral
- 1 jeringa de dosificación oral

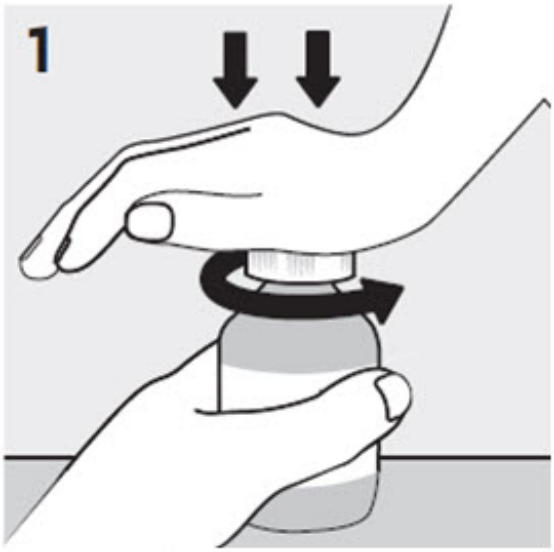


Reúna y verifique los suministros:

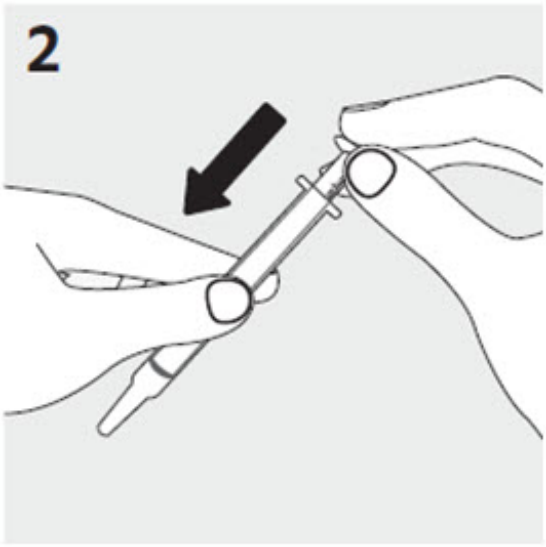
- Reúna el frasco de RISPERDAL Solución oral y la jeringa de dosificación oral.
- Consulte la fecha de caducidad en el frasco. No utilice el frasco de RISPERDAL Solución oral si ha pasado la fecha de caducidad.
- Verifique su dosis en ml según lo prescrito por su proveedor de atención médica. Busque esta marca de ml en el émbolo de la jeringa de dosificación oral. Si su dosis supera los 3 ml, deberá dividir la dosis. Siga las instrucciones que le dé su proveedor de atención médica o farmacéutico sobre cómo dividir su dosis.

Preparación de una dosis de RISPERDAL Solución oral:

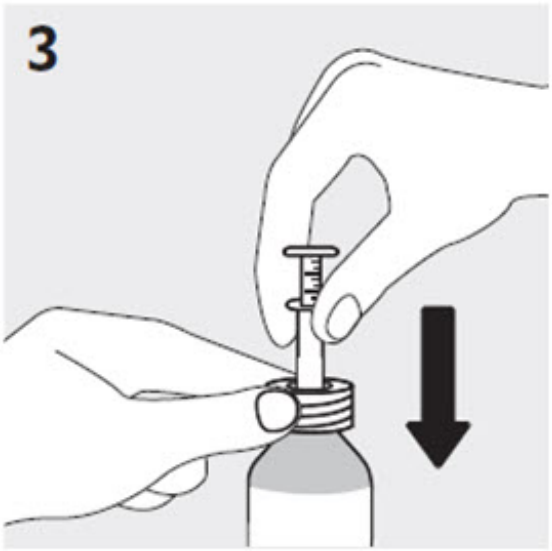
Paso 1. Coloque el frasco de RISPERDAL Solución oral sobre una superficie plana. Empuje hacia abajo la tapa mientras la gira hacia la izquierda (en sentido antihorario) para abrir el frasco.



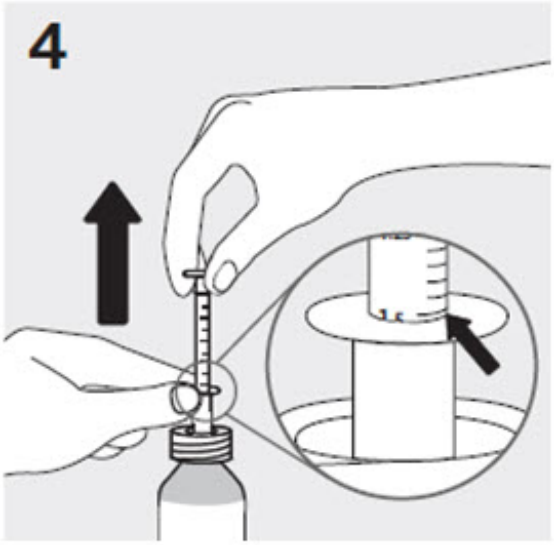
Paso 2. Empuje el émbolo de la jeringa de dosificación oral hasta el fondo.



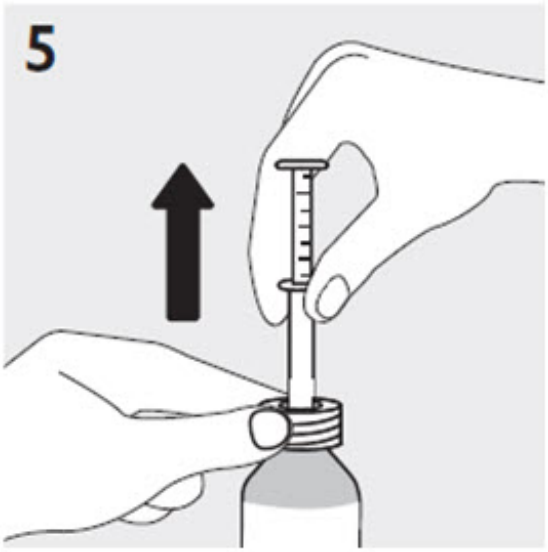
Paso 3. Con el frasco en posición vertical, inserte completamente la jeringa de dosificación oral en la abertura del frasco.



Paso 4. Extraiga la dosis prescrita de RISPERDAL Solución oral del frasco. Sostenga el cilindro de la jeringa de dosificación oral con una mano. Con la otra mano, tire lentamente del émbolo hacia arriba hasta que alcance las marcas de ml en el émbolo para la dosis prescrita.

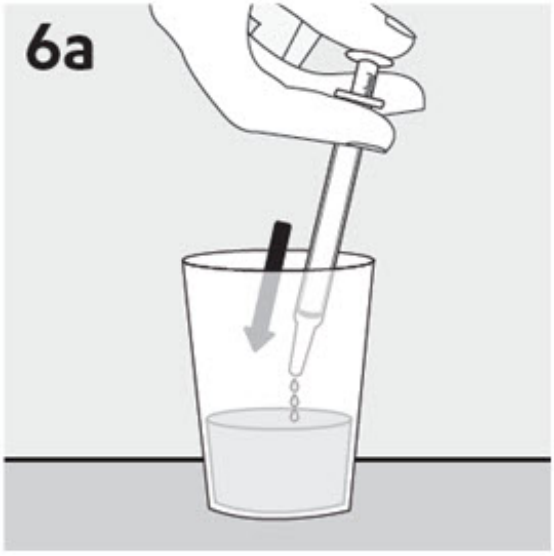


Paso 5. Retire la jeringa de dosificación oral del frasco sujetando el cilindro exterior y tirando hacia arriba. Tenga cuidado de no empujar hacia abajo el émbolo durante este paso.
Compruebe que no haya burbujas de aire en la jeringa de dosificación oral. Si ve burbujas de aire, empuje lentamente el émbolo hasta el fondo para devolver la solución oral al frasco. Luego repita el **Paso 4**para retirar la dosis prescrita.



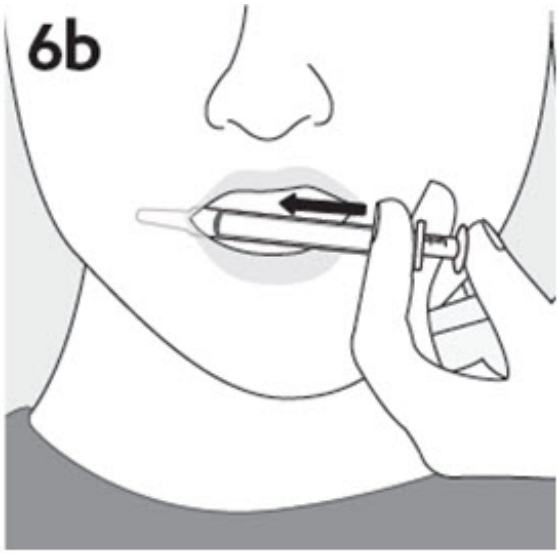
Paso 6.RISPERDAL Solución oral se puede mezclar con una bebida o se puede tomar directamente de la jeringa de dosificación oral.

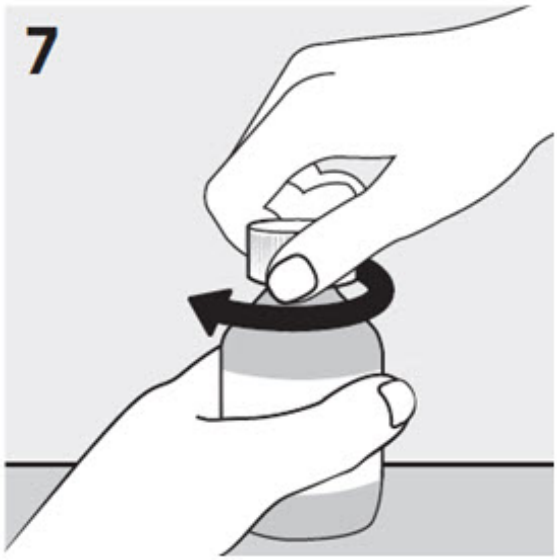
- Mezcle la dosis de RISPERDAL Solución oral con agua, café, jugo de naranja o leche con bajo contenido de grasa. Revuelva bien y beba toda la mezcla de inmediato para asegurarse de tomar la dosis completa. **Consulte la Figura 6a.**
  No mezcleRISPERDAL Solución oral con refrescos de cola o té.



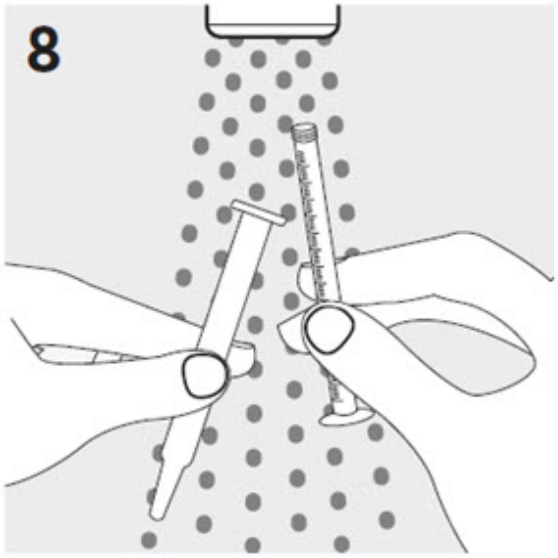
O

- Para tomar la dosis de RISPERDAL Solución oral directamente de la jeringa de dosificación oral, coloque la punta de la jeringa de dosificación oral en la boca y hacia la mejilla. Empuje lentamente el émbolo hasta el fondo para liberar suavemente todo el medicamento en la jeringa de dosificación oral. No rocíe ni empuje con fuerza el medicamento en la parte posterior de la garganta. **Consulte la Figura 6b**.





Paso 7. Vuelva a colocar la tapa en el frasco de RISPERDAL Solución oral y gírela hacia la derecha (en el sentido de las agujas del reloj) para cerrar el frasco.



Paso 8. Enjuague la jeringa de dosificación oral con agua después de cada uso.

- Retire el émbolo del cilindro de la jeringa de dosificación oral.
- Enjuague el cilindro de la jeringa de dosificación oral con agua y déjela secar al aire.
- Cuando el cilindro de la jeringa de dosificación oral y el émbolo estén secos, vuelva a colocar el émbolo en el cilindro de la jeringa de dosificación oral para el próximo uso.

No deseche la jeringa de dosificación oral.

Almacenamiento de RISPERDAL Solución oral

- Almacene RISPERDAL Solución oral a temperatura ambiente entre 59 °F y 77 °F (de 15 °C a 25 °C).
- No congelar RISPERDAL Solución oral. Proteger de la luz.
- **Mantenga RISPERDAL Solución oral y todos los medicamentos fuera del**

alcance de los niños.

Fabricado para:

Janssen Pharmaceuticals, Inc.

Titusville, NJ, 08560, USA

Estas Instrucciones de uso han sido aprobadas por la Administración de Alimentos y Medicamentos (FDA) de los EE. UU.

Para información sobre patentes : www.janssenpatents.com

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Revisado: 2/2025

PRINCIPAL DISPLAY PANEL - 0.5 mg Bottle Label

NDC50458-302-06

RISPERDAL®
(risperiDONE)
TABLETS

0.5 mg

Each tablet contains: Risperidone 0.5 mg

Keep out of reach of children.

Rx only

60 Tablets



PRINCIPAL DISPLAY PANEL - 1 mg Bottle Label

NDC50458-300-06

RISPERDAL®
(risperiDONE)
TABLETS

1 mg

Each tablet contains: Risperidone 1 mg

Keep out of reach of children.

Rx only

60 Tablets



PRINCIPAL DISPLAY PANEL - 2 mg Bottle Label

NDC50458-320-06

RISPERDAL®
(risperiDONE)
TABLETS

2 mg

Each tablet contains: Risperidone 2 mg

Keep out of reach of children.

Rx only

60 Tablets



PRINCIPAL DISPLAY PANEL - 3 mg Bottle Label

60 TABLETS
NDC 50458-330-06

RISPERDAL®
(risperiDONE)
TABLETS

3 mg

Each tablet contains: Risperidone 3 mg

Keep out of reach of children.

Rx only

60 Tablets



PRINCIPAL DISPLAY PANEL - 4 mg Bottle Label

60 TABLETS
NDC 50458-350-06

RISPERDAL®

(risperiDONE)
TABLETS

4 mg

Each tablet contains: Risperidone 4 mg

Keep out of reach of children.

Rx only

60 Tablets



PRINCIPAL DISPLAY PANEL - 0.5 mg Tablet Carton

NDC 50458-395-28
28 TABLETS

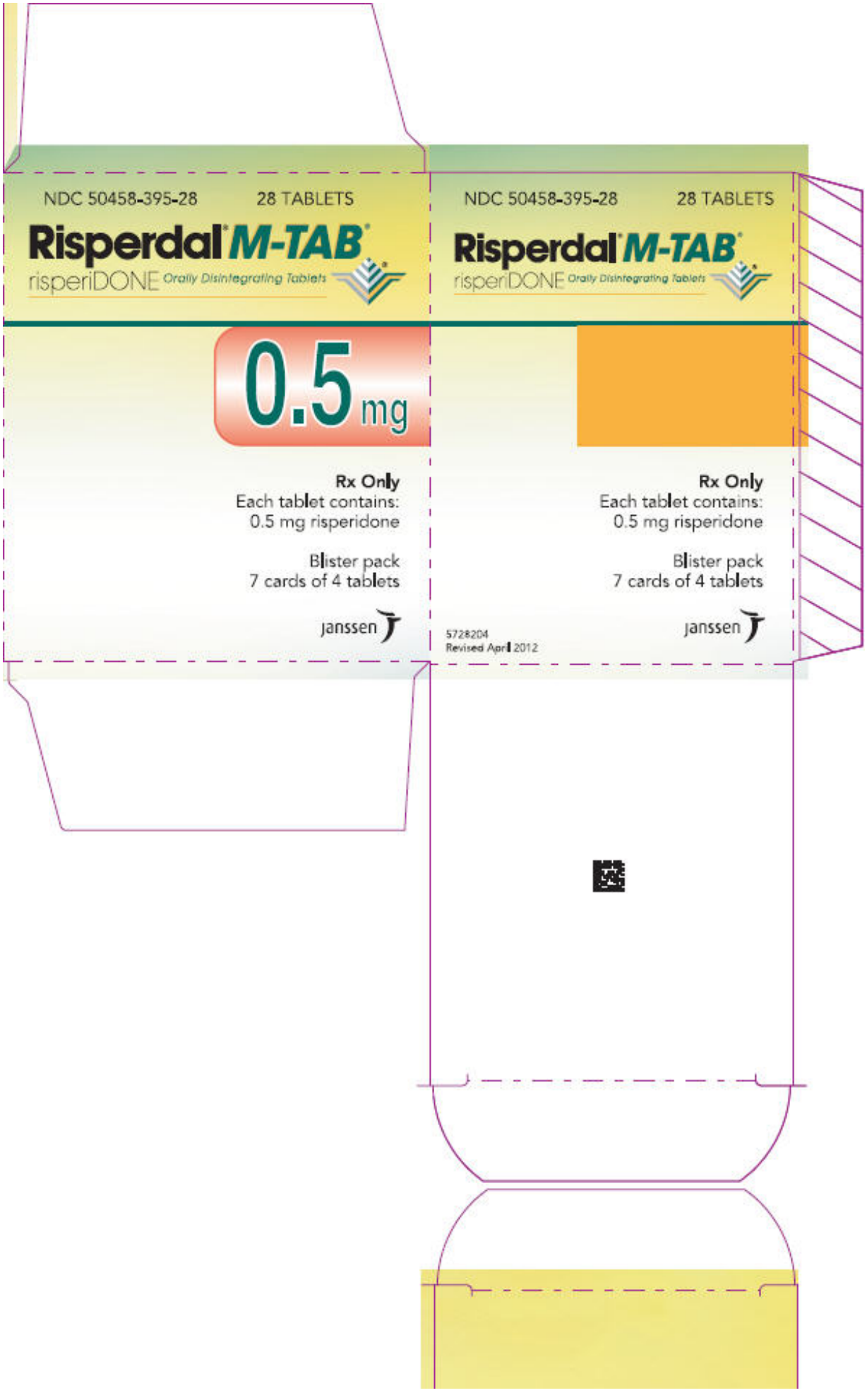
Risperdal®*M-TAB*®
risperiDONE *Orally Disintegrating Tablets*

0.5 mg

Rx Only
Each tablet contains:
0.5 mg risperidone

Blister pack
7 cards of 4 tablets

janssen


NDC 50458-395-28 28 TABLETS
Risperdal M-TAB
risperiDONE Orally Disintegrating Tablets
0.5 mg
Rx Only
Each tablet contains:
0.5 mg risperidone
Blister pack
7 cards of 4 tablets
janssen
5728204
Revised April 2012
NDC 50458-395-28 28 TABLETS
Risperdal M-TAB
risperiDONE Orally Disintegrating Tablets
Rx Only
Each tablet contains:
0.5 mg risperidone
Blister pack
7 cards of 4 tablets
janssen



PRINCIPAL DISPLAY PANEL - 1 mg Tablet Carton

NDC 50458-315-28
28 TABLETS

Risperdal®*M-TAB*®
risperiDONE *Orally Disintegrating Tablets*

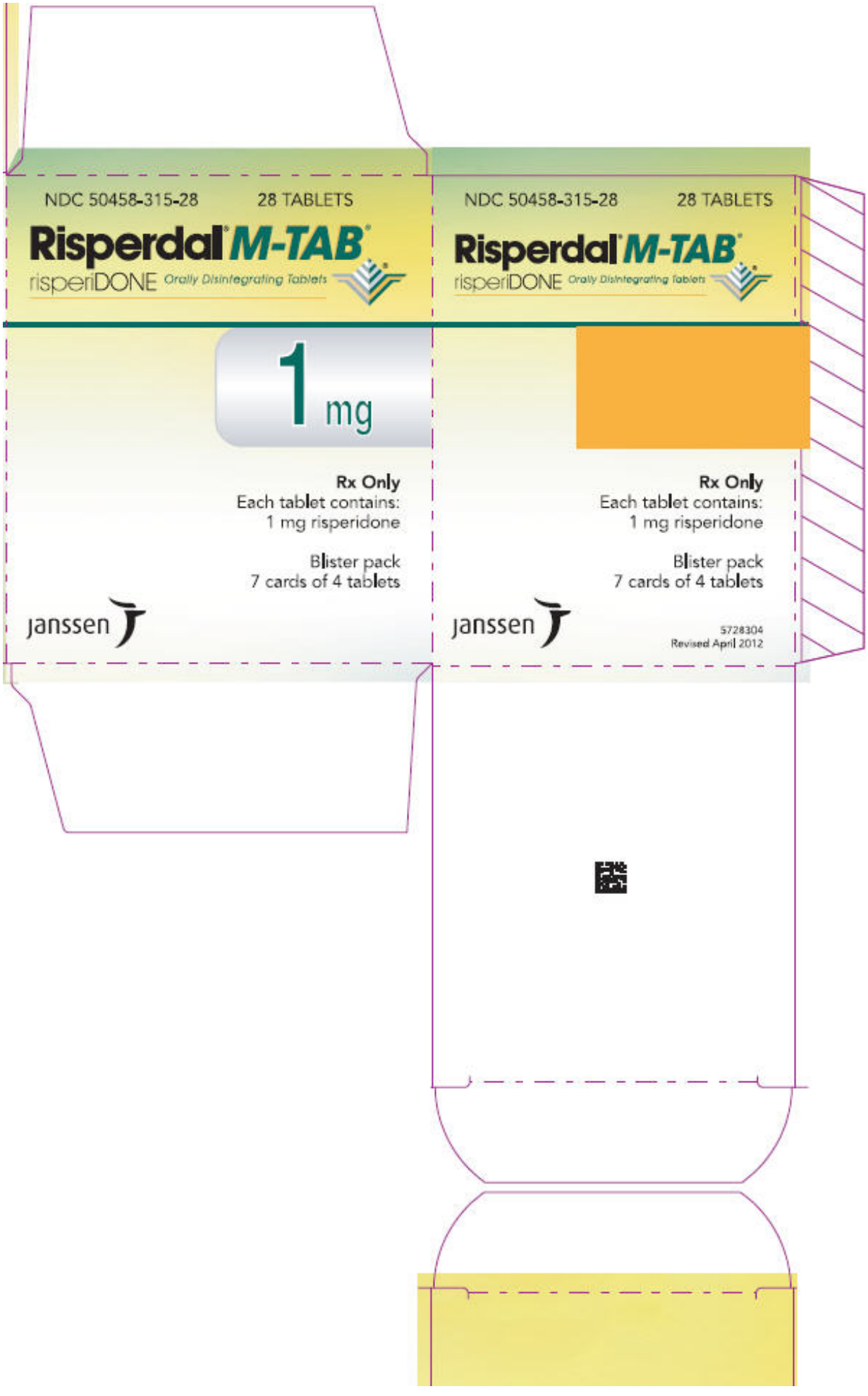
1 mg

Rx Only
Each tablet contains:
1 mg risperidone

Blister pack

7 cards of 4 tablets

janssen


NDC 50458-315-28 28 TABLETS
Risperdal M-TAB
risperiDONE Orally Disintegrating Tablets
1 mg
Rx Only
Each tablet contains:
1 mg risperidone
Blister pack
7 cards of 4 tablets
janssen
NDC 50458-315-28 28 TABLETS
Risperdal M-TAB
risperiDONE Orally Disintegrating Tablets
Rx Only
Each tablet contains:
1 mg risperidone
Blister pack
7 cards of 4 tablets
janssen
5728304
Revised April 2012



PRINCIPAL DISPLAY PANEL - 2 mg Tablet Carton

NDC 50458-325-28
28 TABLETS

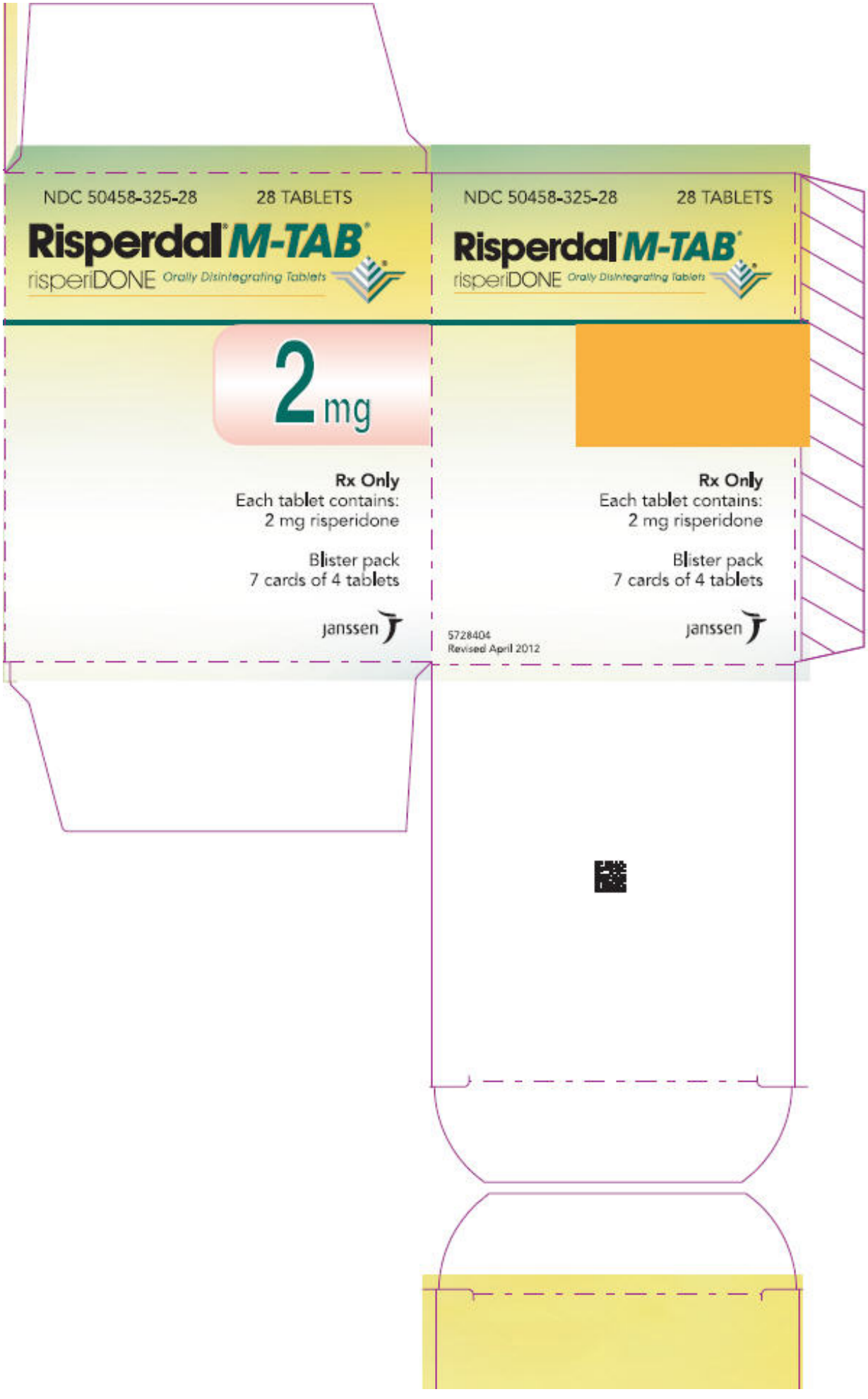
Risperdal®*M-TAB*®
risperiDONE *Orally Disintegrating Tablets*

2 mg

Rx Only
Each tablet contains:
2 mg risperidone

Blister pack
7 cards of 4 tablets

janssen


NDC 50458-325-28 28 TABLETS
Risperdal® M-TAB®
risperiDONE Orally Disintegrating Tablets
2 mg
Rx Only
Each tablet contains:
2 mg risperidone
Blister pack
7 cards of 4 tablets
janssen
5728404
Revised April 2012
NDC 50458-325-28 28 TABLETS
Risperdal® M-TAB®
risperiDONE Orally Disintegrating Tablets
Rx Only
Each tablet contains:
2 mg risperidone
Blister pack
7 cards of 4 tablets
janssen



PRINCIPAL DISPLAY PANEL - 3 mg Tablet Carton

NDC 50458-335-28

Risperdal®*M-TAB*®
risperiDONE *Orally Disintegrating Tablets*

Phenylketonurics: Contains 0.63 mg phenylalanine (a component of aspartame) per 3 mg tablet
Each tablet contains:3 mg risperidone

Rx only

Dosage: For information or use, see accompanying full Prescribing Information.

Product of Ireland

Manufactured by:
Janssen Ortho, LLC
Gurabo, PR 00778

Manufactured for:
Janssen Pharmaceuticals, Inc.
Titusville, NJ 08560

janssen

NDC 50458-335-28

Risperdal® M-TAB®

risperiDONE Orally Disintegrating Tablets

Phenylketonurics: Contains 0.63 mg phenylalanine (a component of aspartame) per 3 mg tablet
Each tablet contains: 3 mg risperidone
Rx only
Dosage: For information or use, see accompanying full Prescribing Information.
Product of Ireland

Manufactured by:
Janssen Ortho, LLC
Gurabo, PR 00778

Manufactured for:
Janssen Pharmaceuticals, Inc.
Titusville, NJ 08560

janssen

3 mg

janssen

Risperdal® M-TAB®
risperiDONE Orally Disintegrating Tablets

NDC 50458-335-28

NDC 50458-335-28

Risperdal® M-TAB®
risperiDONE Orally Disintegrating Tablets

Store at controlled room temperature 15°C–25°C (59°F–77°F).
Protect from light and moisture.
Keep out of reach of children.

3 mg

LOT: and EXP:

© Janssen 2005
Printed in U.S.A.

10074702
Revised April 2012

N 3 50458-335-28 6



PRINCIPAL DISPLAY PANEL - 4 mg Tablet Carton

NDC 50458-355-28

Risperdal®*M-TAB*®
risperiDONE *Orally Disintegrating Tablets*

Phenylketonurics: Contains 0.84 mg phenylalanine (a component of aspartame) per 4 mg tablet
Each tablet contains:4 mg risperidone

Rx only

Dosage: For information or use, see accompanying full Prescribing Information.

Product of Ireland

Manufactured by:
Janssen Ortho, LLC
Gurabo, PR 00778

Manufactured for:

Janssen Pharmaceuticals, Inc.
Titusville, NJ 08560

janssen

NDC 50458-355-28

Risperdal® M-TAB®

risperiDONE Orally Disintegrating Tablets

Phenylketonurics: Contains 0.84 mg phenylalanine (a component of aspartame) per 4 mg tablet
Each tablet contains: 4 mg risperidone
Rx only
Dosage: For information or use, see accompanying full Prescribing Information.
Product of Ireland

Manufactured by:
Janssen Ortho, LLC
Gurabo, PR 00778

Manufactured for:
Janssen Pharmaceuticals, Inc.
Titusville, NJ 08560

janssen

janssen

4 mg

Risperdal® M-TAB®
risperiDONE Orally Disintegrating Tablets

NDC 50458-355-28

NDC 50458-355-28

Risperdal® M-TAB®
risperiDONE Orally Disintegrating Tablets

Store at controlled room temperature 15°C–25°C (59°F–77°F).
Protect from light and moisture.
Keep out of reach of children.

4 mg

LOT, and EXP:

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Printed in U.S.A.

10074802
Revised April 2012

N 3 50458-355-28 4



PRINCIPAL DISPLAY PANEL - 30 mL Bottle Carton

NDC50458-305-03
RISPERDAL®

(risperiDONE)
ORAL SOLUTION

1 mg/mL

Each 1 mL contains 1 mg of risperidone in an aqueous solution.

Rx only

30 mL

Johnson&Johnson



RISPERDAL

risperidone tablet

Product Information			
Product Type	HUMAN PRESCRIPTION DRUG	**Item Code (Source)**	NDC:50458-302
Route of Administration	ORAL		

Active Ingredient/Active Moiety

Ingredient Name	Basis of Strength	Strength
RISPERIDONE (UNII: L6UH7ZF8HC) (RISPERIDONE - UNII:L6UH7ZF8HC)	RISPERIDONE	0.5 mg

Inactive Ingredients

Ingredient Name	Strength
SILICON DIOXIDE (UNII: ETJ7Z6XBU4)	
HYPROMELLOSE, UNSPECIFIED (UNII: 3NXW29V3WO)	
ANHYDROUS LACTOSE (UNII: 3SY5LH9PMK)	
MAGNESIUM STEARATE (UNII: 70097M6I30)	
MICROCRYSTALLINE CELLULOSE (UNII: OP1R32D61U)	
PROPYLENE GLYCOL (UNII: 6DC9Q167V3)	
SODIUM LAURYL SULFATE (UNII: 368GB5141J)	
STARCH, CORN (UNII: O8232NY3SJ)	
TALC (UNII: 7SEV7J4R1U)	
TITANIUM DIOXIDE (UNII: 15FIX9V2JP)	
FERRIC OXIDE RED (UNII: 1K09F3G675)	

Product Characteristics

Color	BROWN (red brown)	**Score**	no score
Shape	OVAL (capsule-shaped)	**Size**	9mm
Flavor		**Imprint Code**	Ris;0;5;JANSSEN
Contains			

Packaging

#	Item Code	Package Description	Marketing Start Date	Marketing End Date
1	NDC:50458-302-06	60 in 1 BOTTLE; Type 0: Not a Combination Product	12/29/1993	
2	NDC:50458-302-50	500 in 1 BOTTLE; Type 0: Not a Combination Product	12/29/1993	02/29/2020
3	NDC:50458-302-01	10 in 1 BOX, UNIT-DOSE	12/29/1993	05/30/2026
3		10 in 1 BLISTER PACK; Type 0: Not a Combination Product		

Marketing Information

Marketing Category	Application Number or Monograph Citation	Marketing Start Date	Marketing End Date
NDA	NDA020272	12/29/1993	

RISPERDAL

risperidone tablet

Product Information

Product Type	HUMAN PRESCRIPTION DRUG	**Item Code (Source)**	NDC:50458-300
Route of Administration	ORAL		

Active Ingredient/Active Moiety

Ingredient Name	Basis of Strength	Strength
RISPERIDONE (UNII: L6UH7ZF8HC) (RISPERIDONE - UNII:L6UH7ZF8HC)	RISPERIDONE	1 mg

Inactive Ingredients

Ingredient Name	Strength
SILICON DIOXIDE (UNII: ETJ7Z6XBU4)	
HYPROMELLOSE, UNSPECIFIED (UNII: 3NXW29V3WO)	
ANHYDROUS LACTOSE (UNII: 3SY5LH9PMK)	
MAGNESIUM STEARATE (UNII: 70097M6I30)	
MICROCRYSTALLINE CELLULOSE (UNII: OP1R32D61U)	
PROPYLENE GLYCOL (UNII: 6DC9Q167V3)	
SODIUM LAURYL SULFATE (UNII: 368GB5141J)	
STARCH, CORN (UNII: O8232NY3SJ)	

Product Characteristics

Color	white	**Score**	no score
Shape	OVAL (capsule-shaped)	**Size**	12mm
Flavor		**Imprint Code**	R1;JANSSEN
Contains			

Packaging

#	Item Code	Package Description	Marketing Start Date	Marketing End Date
1	NDC:50458-300-06	60 in 1 BOTTLE; Type 0: Not a Combination Product	12/29/1993	
2	NDC:50458-300-50	500 in 1 BOTTLE; Type 0: Not a Combination Product	12/29/1993	01/31/2020
3	NDC:50458-300-01	10 in 1 BOX, UNIT-DOSE	12/29/1993	
3		10 in 1 BLISTER PACK; Type 0: Not a Combination Product		

Marketing Information

Marketing Category	Application Number or Monograph Citation	Marketing Start Date	Marketing End Date
NDA	NDA020272	12/29/1993	

RISPERDAL

risperidone tablet

Product Information			
Product Type	HUMAN PRESCRIPTION DRUG	**Item Code (Source)**	NDC:50458-320
Route of Administration	ORAL		

Active Ingredient/Active Moiety		
Ingredient Name	**Basis of Strength**	**Strength**
RISPERIDONE (UNII: L6UH7ZF8HC) (RISPERIDONE - UNII:L6UH7ZF8HC)	RISPERIDONE	2 mg

Inactive Ingredients	
Ingredient Name	**Strength**
SILICON DIOXIDE (UNII: ETJ7Z6XBU4)	
HYPROMELLOSE, UNSPECIFIED (UNII: 3NXW29V3WO)	
ANHYDROUS LACTOSE (UNII: 3SY5LH9PMK)	
MAGNESIUM STEARATE (UNII: 70097M6I30)	
MICROCRYSTALLINE CELLULOSE (UNII: OP1R32D61U)	
PROPYLENE GLYCOL (UNII: 6DC9Q167V3)	
SODIUM LAURYL SULFATE (UNII: 368GB5141J)	
STARCH, CORN (UNII: O8232NY3SJ)	
TALC (UNII: 7SEV7J4R1U)	
TITANIUM DIOXIDE (UNII: 15FIX9V2JP)	
FD&C YELLOW NO. 6 (UNII: H77VEI93A8)	
ALUMINUM OXIDE (UNII: LMI26O6933)	

Product Characteristics			
Color	ORANGE	**Score**	no score
Shape	OVAL (capsule-shaped)	**Size**	12mm
Flavor		**Imprint Code**	R2;JANSSEN
Contains			

Packaging				
#	**Item Code**	**Package Description**	**Marketing Start Date**	**Marketing End Date**
1	NDC:50458-320-06	60 in 1 BOTTLE; Type 0: Not a Combination Product	12/29/1993	
2	NDC:50458-320-50	500 in 1 BOTTLE; Type 0: Not a Combination Product	12/29/1993	10/31/2019
3	NDC:50458-320-01	10 in 1 BOX, UNIT-DOSE	12/29/1993	08/31/2026
3		10 in 1 BLISTER PACK; Type 0: Not a Combination Product		

Marketing Information

Marketing Category	Application Number or Monograph Citation	Marketing Start Date	Marketing End Date
NDA	NDA020272	12/29/1993	

RISPERDAL

risperidone tablet

Product Information

Product Type	HUMAN PRESCRIPTION DRUG	**Item Code (Source)**	NDC:50458-330
Route of Administration	ORAL		

Active Ingredient/Active Moiety

Ingredient Name	Basis of Strength	Strength
RISPERIDONE (UNII: L6UH7ZF8HC) (RISPERIDONE - UNII:L6UH7ZF8HC)	RISPERIDONE	3 mg

Inactive Ingredients

Ingredient Name	Strength
SILICON DIOXIDE (UNII: ETJ7Z6XBU4)	
HYPROMELLOSE, UNSPECIFIED (UNII: 3NXW29V3WO)	
ANHYDROUS LACTOSE (UNII: 3SY5LH9PMK)	
MAGNESIUM STEARATE (UNII: 70097M6I30)	
MICROCRYSTALLINE CELLULOSE (UNII: OP1R32D61U)	
PROPYLENE GLYCOL (UNII: 6DC9Q167V3)	
SODIUM LAURYL SULFATE (UNII: 368GB5141J)	
STARCH, CORN (UNII: O8232NY3SJ)	
TALC (UNII: 7SEV7J4R1U)	
TITANIUM DIOXIDE (UNII: 15FIX9V2JP)	
D&C YELLOW NO. 10 (UNII: 35SW5USQ3G)	

Product Characteristics

Color	yellow	**Score**	no score
Shape	OVAL (capsule-shaped)	**Size**	14mm
Flavor		**Imprint Code**	R3;JANSSEN
Contains			

Packaging

#	Item Code	Package Description	Marketing Start Date	Marketing End Date
1	NDC:50458-330-06	60 in 1 BOTTLE; Type 0: Not a Combination Product	12/29/1993	
2	NDC:50458-330-50	500 in 1 BOTTLE; Type 0: Not a Combination Product	12/29/1993	04/30/2019

3	NDC:50458-330-01	10 in 1 BOX, UNIT-DOSE	12/29/1993	11/07/2025
3		10 in 1 BLISTER PACK; Type 0: Not a Combination Product		

Marketing Information

Marketing Category	Application Number or Monograph Citation	Marketing Start Date	Marketing End Date
NDA	NDA020272	12/29/1993	

RISPERDAL

risperidone tablet

Product Information

Product Type	HUMAN PRESCRIPTION DRUG	**Item Code (Source)**	NDC:50458-350
Route of Administration	ORAL		

Active Ingredient/Active Moiety

Ingredient Name	Basis of Strength	Strength
RISPERIDONE (UNII: L6UH7ZF8HC) (RISPERIDONE - UNII:L6UH7ZF8HC)	RISPERIDONE	4 mg

Inactive Ingredients

Ingredient Name	Strength
SILICON DIOXIDE (UNII: ETJ7Z6XBU4)	
HYPROMELLOSE, UNSPECIFIED (UNII: 3NXW29V3WO)	
ANHYDROUS LACTOSE (UNII: 3SY5LH9PMK)	
MAGNESIUM STEARATE (UNII: 70097M6I30)	
MICROCRYSTALLINE CELLULOSE (UNII: OP1R32D61U)	
PROPYLENE GLYCOL (UNII: 6DC9Q167V3)	
SODIUM LAURYL SULFATE (UNII: 368GB5141J)	
STARCH, CORN (UNII: O8232NY3SJ)	
TALC (UNII: 7SEV7J4R1U)	
TITANIUM DIOXIDE (UNII: 15FIX9V2JP)	
D&C YELLOW NO. 10 (UNII: 35SW5USQ3G)	
FD&C BLUE NO. 2 (UNII: L06K8R7DQK)	
ALUMINUM OXIDE (UNII: LMI26O6933)	

Product Characteristics

Color	GREEN	**Score**	no score
Shape	OVAL (capsule-shaped)	**Size**	15mm
Flavor		**Imprint Code**	R4;JANSSEN
Contains			

Packaging				
#	Item Code	Package Description	Marketing Start Date	Marketing End Date
1	NDC:50458-350-06	60 in 1 BOTTLE; Type 0: Not a Combination Product	12/29/1993	
2	NDC:50458-350-01	10 in 1 BOX, UNIT-DOSE	12/29/1993	08/31/2026
2		10 in 1 BLISTER PACK; Type 0: Not a Combination Product		

Marketing Information

Marketing Category	Application Number or Monograph Citation	Marketing Start Date	Marketing End Date
NDA	NDA020272	12/29/1993	

RISPERDAL M-TAB

risperidone tablet, orally disintegrating

Product Information			
Product Type	HUMAN PRESCRIPTION DRUG	**Item Code (Source)**	NDC:50458-395
Route of Administration	ORAL		

Active Ingredient/Active Moiety		
Ingredient Name	**Basis of Strength**	**Strength**
RISPERIDONE (UNII: L6UH7ZF8HC) (RISPERIDONE - UNII:L6UH7ZF8HC)	RISPERIDONE	0.5 mg

Inactive Ingredients	
Ingredient Name	**Strength**
GELATIN, UNSPECIFIED (UNII: 2G86QN327L)	
MANNITOL (UNII: 3OWL53L36A)	
GLYCINE (UNII: TE7660XO1C)	
CARBOMER HOMOPOLYMER TYPE C (ALLYL PENTAERYTHRITOL CROSSLINKED) (UNII: 4Q93RCW27E)	
SODIUM HYDROXIDE (UNII: 55X04QC32I)	
ASPARTAME (UNII: Z0H242BBR1)	
FERRIC OXIDE RED (UNII: 1K09F3G675)	
PEPPERMINT OIL (UNII: AV092KU4JH)	

Product Characteristics			
Color	orange (light coral)	**Score**	no score
Shape	ROUND (biconvex)	**Size**	10mm
Flavor	PEPPERMINT	**Imprint Code**	R0;5

Contains			

Packaging

#	Item Code	Package Description	Marketing Start Date	Marketing End Date
1	NDC:50458-395-28	7 in 1 BOX	04/02/2003	05/31/2018
1		4 in 1 BLISTER PACK; Type 0: Not a Combination Product		
2	NDC:50458-395-30	30 in 1 BLISTER PACK; Type 0: Not a Combination Product	04/02/2003	05/30/2017

Marketing Information

Marketing Category	Application Number or Monograph Citation	Marketing Start Date	Marketing End Date
NDA	NDA021444	04/02/2003	

RISPERDAL M-TAB

risperidone tablet, orally disintegrating

Product Information

Product Type	HUMAN PRESCRIPTION DRUG	Item Code (Source)	NDC:50458-315
Route of Administration	ORAL		

Active Ingredient/Active Moiety

Ingredient Name	Basis of Strength	Strength
RISPERIDONE (UNII: L6UH7ZF8HC) (RISPERIDONE - UNII:L6UH7ZF8HC)	RISPERIDONE	1 mg

Inactive Ingredients

Ingredient Name	Strength
GELATIN, UNSPECIFIED (UNII: 2G86QN327L)	
MANNITOL (UNII: 3OWL53L36A)	
GLYCINE (UNII: TE7660XO1C)	
CARBOMER HOMOPOLYMER TYPE C (ALLYL PENTAERYTHRITOL CROSSLINKED) (UNII: 4Q93RCW27E)	
SODIUM HYDROXIDE (UNII: 55X04QC32I)	
ASPARTAME (UNII: Z0H242BBR1)	
FERRIC OXIDE RED (UNII: 1K09F3G675)	
PEPPERMINT OIL (UNII: AV092KU4JH)	

Product Characteristics

Color	orange (light coral)	Score	no score

Shape	SQUARE (biconvex)	**Size**	11mm
Flavor	PEPPERMINT	**Imprint Code**	R1
Contains			

Packaging

#	Item Code	Package Description	Marketing Start Date	Marketing End Date
1	NDC:50458-315-28	7 in 1 BOX	04/02/2003	02/28/2018
1		4 in 1 BLISTER PACK; Type 0: Not a Combination Product		
2	NDC:50458-315-30	30 in 1 BLISTER PACK; Type 0: Not a Combination Product	04/02/2003	02/28/2018

Marketing Information

Marketing Category	Application Number or Monograph Citation	Marketing Start Date	Marketing End Date
NDA	NDA021444	04/02/2003	

RISPERDAL M-TAB

risperidone tablet, orally disintegrating

Product Information

Product Type	HUMAN PRESCRIPTION DRUG	**Item Code (Source)**	NDC:50458-325
Route of Administration	ORAL		

Active Ingredient/Active Moiety

Ingredient Name	Basis of Strength	Strength
RISPERIDONE (UNII: L6UH7ZF8HC) (RISPERIDONE - UNII:L6UH7ZF8HC)	RISPERIDONE	2 mg

Inactive Ingredients

Ingredient Name	Strength
GELATIN, UNSPECIFIED (UNII: 2G86QN327L)	
MANNITOL (UNII: 3OWL53L36A)	
GLYCINE (UNII: TE7660XO1C)	
CARBOMER HOMOPOLYMER TYPE C (ALLYL PENTAERYTHRITOL CROSSLINKED) (UNII: 4Q93RCW27E)	
SODIUM HYDROXIDE (UNII: 55X04QC32I)	
ASPARTAME (UNII: Z0H242BBR1)	
FERRIC OXIDE RED (UNII: 1K09F3G675)	
PEPPERMINT OIL (UNII: AV092KU4JH)	
XANTHAN GUM (UNII: TTV12P4NEE)	

Product Characteristics			
Color	orange (coral)	**Score**	no score
Shape	SQUARE (biconvex)	**Size**	11mm
Flavor	PEPPERMINT	**Imprint Code**	R2
Contains			

Packaging				
#	**Item Code**	**Package Description**	**Marketing Start Date**	**Marketing End Date**
1	NDC:50458-325-28	7 in 1 BOX	04/02/2003	09/30/2017
1		4 in 1 BLISTER PACK; Type 0: Not a Combination Product		

Marketing Information

Marketing Category	Application Number or Monograph Citation	Marketing Start Date	Marketing End Date
NDA	NDA021444	04/02/2003	

RISPERDAL M-TAB

risperidone tablet, orally disintegrating

Product Information			
Product Type	HUMAN PRESCRIPTION DRUG	**Item Code (Source)**	NDC:50458-335
Route of Administration	ORAL		

Active Ingredient/Active Moiety		
Ingredient Name	**Basis of Strength**	**Strength**
RISPERIDONE (UNII: L6UH7ZF8HC) (RISPERIDONE - UNII:L6UH7ZF8HC)	RISPERIDONE	3 mg

Inactive Ingredients	
Ingredient Name	**Strength**
GELATIN, UNSPECIFIED (UNII: 2G86QN327L)	
MANNITOL (UNII: 3OWL53L36A)	
GLYCINE (UNII: TE7660XO1C)	
CARBOMER HOMOPOLYMER TYPE C (ALLYL PENTAERYTHRITOL CROSSLINKED) (UNII: 4Q93RCW27E)	
SODIUM HYDROXIDE (UNII: 55X04QC32I)	
ASPARTAME (UNII: Z0H242BBR1)	
FERRIC OXIDE RED (UNII: 1K09F3G675)	
PEPPERMINT OIL (UNII: AV092KU4JH)	
XANTHAN GUM (UNII: TTV12P4NEE)	

Product Characteristics			
Color	orange (coral)	**Score**	no score
Shape	ROUND (biconvex)	**Size**	14mm
Flavor	PEPPERMINT	**Imprint Code**	R3
Contains			

Packaging				
#	**Item Code**	**Package Description**	**Marketing Start Date**	**Marketing End Date**
1	NDC:50458-335-28	7 in 1 BOX	04/02/2003	03/31/2018
1		4 in 1 BLISTER PACK; Type 0: Not a Combination Product		

Marketing Information

Marketing Category	Application Number or Monograph Citation	Marketing Start Date	Marketing End Date
NDA	NDA021444	04/02/2003	

RISPERDAL M-TAB

risperidone tablet, orally disintegrating

Product Information			
Product Type	HUMAN PRESCRIPTION DRUG	**Item Code (Source)**	NDC:50458-355
Route of Administration	ORAL		

Active Ingredient/Active Moiety		
Ingredient Name	**Basis of Strength**	**Strength**
RISPERIDONE (UNII: L6UH7ZF8HC) (RISPERIDONE - UNII:L6UH7ZF8HC)	RISPERIDONE	4 mg

Inactive Ingredients	
Ingredient Name	**Strength**
GELATIN, UNSPECIFIED (UNII: 2G86QN327L)	
MANNITOL (UNII: 3OWL53L36A)	
GLYCINE (UNII: TE7660XO1C)	
CARBOMER HOMOPOLYMER TYPE C (ALLYL PENTAERYTHRITOL CROSSLINKED) (UNII: 4Q93RCW27E)	
SODIUM HYDROXIDE (UNII: 55X04QC32I)	
ASPARTAME (UNII: Z0H242BBR1)	
FERRIC OXIDE RED (UNII: 1K09F3G675)	
PEPPERMINT OIL (UNII: AV092KU4JH)	

XANTHAN GUM (UNII: TTV12P4NEE)	

Product Characteristics			
Color	orange (coral)	**Score**	no score
Shape	ROUND (biconvex)	**Size**	16mm
Flavor	PEPPERMINT	**Imprint Code**	R4
Contains			

Packaging

#	Item Code	Package Description	Marketing Start Date	Marketing End Date
1	NDC:50458-355-28	7 in 1 BOX	04/02/2003	03/31/2018
1		4 in 1 BLISTER PACK; Type 0: Not a Combination Product		

Marketing Information

Marketing Category	Application Number or Monograph Citation	Marketing Start Date	Marketing End Date
NDA	NDA021444	04/02/2003	

RISPERDAL

risperidone solution

Product Information			
Product Type	HUMAN PRESCRIPTION DRUG	**Item Code (Source)**	NDC:50458-305
Route of Administration	ORAL		

Active Ingredient/Active Moiety

Ingredient Name	Basis of Strength	Strength
RISPERIDONE (UNII: L6UH7ZF8HC) (RISPERIDONE - UNII:L6UH7ZF8HC)	RISPERIDONE	1 mg in 1 mL

Inactive Ingredients

Ingredient Name	Strength
TARTARIC ACID (UNII: W4888I119H)	
BENZOIC ACID (UNII: 8SKN0B0MIM)	
SODIUM HYDROXIDE (UNII: 55X04QC32I)	
WATER (UNII: 059QF0KO0R)	

Packaging

#	Item Code	Package Description	Marketing Start Date	Marketing End Date
1	NDC:50458-305-03	30 mL in 1 BOTTLE; Type 0: Not a Combination Product	06/10/1996	

Marketing Information

Marketing Category	Application Number or Monograph Citation	Marketing Start Date	Marketing End Date
NDA	NDA020588	06/10/1996	

Labeler - Janssen Pharmaceuticals, Inc. (063137772)

Establishment

Name	Address	ID/FEI	Business Operations
Janssen Pharmaceutica NV		370005019	ANALYSIS(50458-305) , MANUFACTURE(50458-305)

Establishment

Name	Address	ID/FEI	Business Operations
Janssen Pharmaceutica NV		400345889	API MANUFACTURE(50458-305)

Establishment

Name	Address	ID/FEI	Business Operations
Janssen Ortho LLC		805887986	ANALYSIS(50458-300, 50458-302, 50458-315, 50458-320, 50458-325, 50458-330, 50458-335, 50458-350, 50458-355, 50458-395) , MANUFACTURE(50458-300, 50458-302, 50458-315, 50458-320, 50458-325, 50458-330, 50458-335, 50458-350, 50458-355, 50458-395)

Establishment

Name	Address	ID/FEI	Business Operations
Janssen Pharmaceutical Sciences Unlimited Company		985639841	API MANUFACTURE(50458-300, 50458-302, 50458-305, 50458-315, 50458-320, 50458-325, 50458-330, 50458-335, 50458-350, 50458-355, 50458-395)

Establishment

Name	Address	ID/FEI	Business Operations
LUSOMEDICAMENTA - SOCIEDADE TÉCNICA FARMACÊUTICA, S.A.		450630343	ANALYSIS(50458-305) , MANUFACTURE(50458-305) , PACK(50458-305) , LABEL(50458-305)

Revised: 10/2026 Janssen Pharmaceuticals, Inc.